3

KINEMATICS OF FLUID FLOW

3.1 INTRODUCTION

Kinematics of fluid flow is that branch of fluid mechanics which describes the fluid motion and its consequences without consideration of the nature of forces causing the motion. The basic understanding of the fluid kinematics forms the groundwork for the studies on dynamical behaviour of fluids in consideration of the forces accompanying the motion. The subject has the following three main aspects:

(a) The development of methods and techniques for describing and specifying the motions of fluids.
(b) Characterisation of different types of motion and associated deformation rates of any fluid element.
(c) The determination of the conditions for the kinematic possibility of fluid motions, i.e., the exploration of the consequences of continuity in the motion.

3.2 SCALAR AND VECTOR FIELDS

Scalar A quantity which has only magnitude is defined to be a scalar. A scalar quantity can be completely specified by a single number representing its magnitude. Typical scalar quantities are mass, density and temperature. The magnitude of a scalar (a real number) will change when the units expressing the scalar are changed, but the physical entity remains the same.

Vector A quantity which is specified by both magnitude and direction is known to be a vector. Force, velocity and displacement are typical vector quantities. The magnitude of a vector is a scalar.

Scalar Field If at every point in a region, a scalar function has a defined value, the region is called a scalar field. The temperature distribution in a rod is an example of a scalar field.

Vector Field If at every point in a region, a vector function has a defined value, the region is called a vector field. Force and velocity fields are typical examples of vector fields.

3.3 FLOW FIELD AND DESCRIPTION OF FLUID MOTION

A flow field is a region in which the flow velocity is defined at each and every point in space, at any instant of time. Usually, the velocity describes the flow. In other words, a flow field is specified by the velocities at different points in the region at different times. A fluid mass can be conceived of consisting of a number of fluid particles. Hence the instantaneous velocity at any point in a fluid region is actually the velocity of a particle (of the same density as that of the fluid) that exists at that point at that instant. In order to obtain a complete picture of the flow, the fluid motion is described by two methods discussed as follows:

1. *Lagrangian Method* In this method, the fluid motion is described by tracing the kinematic behaviour of each and every individual particle constituting the flow. Identities of the particles are made by specifying their initial position (spatial location) at a given time. The position of a particle at any other instant of time then becomes a function of its identity and time. This statement can be analytically expressed as

$$\vec{S} = S(\vec{S}_0, t) \tag{3.1}$$

where $\vec{S}$ is the position vector of a particle (with respect to a fixed point of reference) at a time t. $\vec{S}_0$ is its initial position at a given time $t = t_0$, and thus specifies the identity of the particle. Equation (3.1) can be written into scalar components with respect to a rectangular Cartesian frame of coordinates as

$$x = x(x_0, y_0, z_0, t) \tag{3.1a}$$

$$y = y(x_0, y_0, z_0, t) \tag{3.1b}$$

$$z = z(x_0, y_0, z_0, t) \tag{3.1c}$$

where x_0, y_0, z_0 are the initial coordinates and x, y, z are the coordinates at a time t of the particle. Hence $\vec{S}$ in Eq. (3.1) can be expressed as

$$\vec{S} = \hat{i}\,x + \hat{j}\,y + \hat{k}\,z$$

where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors along the x, y and z axes, respectively. The velocity $\vec{V}$ and acceleration $\vec{a}$ of the fluid particle can be obtained from the material derivatives of the position of the particle with respect to time.
Therefore,

$$\vec{V} = \left[\frac{\mathrm{d}\vec{S}}{\mathrm{d}t}\right]_{S_0} \tag{3.2a}$$

or, in terms of scalar components,

$$u = \left[\frac{\mathrm{d}x}{\mathrm{d}t}\right]_{x_0, y_0 z_0} \tag{3.2b}$$

$$v = \left[\frac{\mathrm{d}y}{\mathrm{d}t}\right]_{x_0, y_0 z_0} \tag{3.2c}$$

$$w = \left[\frac{\mathrm{d}z}{\mathrm{d}t}\right]_{x_0,\, y_0,\, z_0} \tag{3.2d}$$

u, v, w are the components of velocity in x, y and z directions, respectively. For the acceleration,

$$\vec{a} = \left[\frac{\mathrm{d}^2\vec{S}}{\mathrm{d}t^2}\right]_{S_0} \tag{3.3a}$$

and hence,

$$a_x = \left[\frac{\mathrm{d}^2x}{\mathrm{d}t^2}\right]_{x_0, y_0, z_0} \tag{3.3b}$$

$$a_y = \left[\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right]_{x_0, y_0, z_0} \tag{3.3c}$$

$$a_z = \left[\frac{\mathrm{d}^2z}{\mathrm{d}t^2}\right]_{x_0, y_0, z_0} \tag{3.3d}$$

The subscripts in Eqs (3.2) and (3.3) represent the initial (at $t = t_0$) position of the particle and thus specify the particle identity. The favourable aspect of the method lies in the information about the motion and trajectory of each and every particle of the fluid so that at any time it is possible to trace the history of each fluid particle. In addition, by virtue of the fact that particles are initially identified and traced through their motion, conservation of mass is inherent. However, the serious drawback of this method is that the solution of the equations (Eqs (3.2) and (3.3)) presents appreciable mathematical difficulties except certain special cases and therefore, the method is rarely suitable for practical applications.

2. *Eulerian Method* The method given by Leonhard Euler is of greater advantage since it avoids the determination of the movement of each individual fluid particle in all details. Instead it seeks the velocity $\vec{V}$ and its variation with time t at each and every location ($\vec{S}$) in the flow field. In the Lagrangian view, all hydrodynamic parameters are tied to the particles or elements, whereas in the Eulerian view, they are functions of location and time. Mathematically, the flow field in the Eulerian method is described as

$$\vec{V} = \vec{V}(\vec{S}, t) \tag{3.4}$$

where, $\vec{V} = \hat{i}u + \hat{j}v + \hat{k}w$

and $\vec{S} = \hat{i}x + \hat{j}y + \hat{k}z$

Therefore,

$$u = u(x, y, z, t) \tag{3.4a}$$

$$v = v(x, y, z, t) \tag{3.4b}$$

$$w = w(x, y, z, t) \tag{3.4c}$$

The relationship between the Eulerian and the Lagrangian method can now be shown. Equation (3.4) of Eulerian description can be written as

$$\frac{d\vec{S}}{dt} = \vec{V}(\vec{S}, t) \tag{3.5}$$

or

$$\frac{dx}{dt} = u(x, y, z, t) \tag{3.5a}$$

$$\frac{dy}{dt} = v(x, y, z, t) \tag{3.5b}$$

$$\frac{dz}{dt} = w(x, y, z, t) \tag{3.5c}$$

The integration of Eq. (3.5) yields the constants of integration which are to be found from the initial coordinates of the fluid particles. Hence, the solution of Eq. (3.5) gives the equations of Lagrange as

$$\vec{S} = \vec{S}(\vec{S}_0, t)$$

or

$$x = x(x_0, y_0, z_0, t)$$
$$y = y(x_0, y_0, z_0, t)$$
$$z = z(x_0, y_0, z_0, t)$$

Therefore, it is evident that, in principle, the Lagrangian method of description can always be derived from the Eulerian method. But the solution of the set of three simultaneous differential equations is generally very difficult.

Example 3.1

In a one-dimensional flow field, the velocity at a point may be given in the Eulerian system by $u = x + t$. Determine the displacement of a fluid particle whose initial position is x_0 at initial time t_0 in the Lagrangian system.

Solution

$$u = x + t$$

or

$$dx/dt = x + t$$

Using D as the operator d/dt, the above equation can be written as

$$(D - 1)x = t$$

The solution of the equation is

$$x = Ae^t - t - 1$$

The constant A is found from the initial condition as follows:

$$x_0 = Ae^{t0} - t_0 - 1$$

Hence,

$$A = \frac{x_0 + t_0 + 1}{e^{t_0}}$$

Substituting the value of A in the solution, we get

$$x = (x_0 + t_0 + 1)\, e^{(t - t0)} - t - 1$$

This equation is the required Lagrangian version of the fluid particle having the identity $x = x_0$ at $t = t_0$.

Example 3.2

A two-dimensional flow is described in the Lagrangian system as

$$x = x_0 e^{-kt} + y_0 (1 - e^{-2kt})$$

and

$$y = y_0 e^{kt}$$

Find (a) the equation of a fluid particle in the flow field and (b) the velocity components in the Eulerian system.

Solution

(a) Trajectory of fluid particle in the flow field is found by eliminating t from the equations describing its motion as follows:

$$e^{kt} = y/y_0$$

Hence,

$$x = x_0(y_0/y) + y_0(1 - y_0^2/y^2)$$

which finally gives after some arrangement

$$(x - y_0)y^2 - x_0 y_0 y + y_0^3 = 0$$

This is the required equation

(b) u (the x component of velocity)

$$= \frac{dx}{dt}$$

$$= \frac{d}{dt}\left[x_0 e^{-kt} + y_0(1 - e^{-2kt})\right]$$

$$= -k x_0 e^{-kt} + 2 k y_0 e^{-2kt}$$

$$= -k\,[x - y_0(1 - e^{-2kt})] + 2 k y_0 e^{-2kt}$$

$$= -kx + k y_0 (1 + e^{-2kt})$$

$$= -kx + ky\,(e^{-kt} + e^{-3kt})$$

v (the y component of velocity)

$$= \frac{dy}{dt} = \frac{d}{dt}\left(y_0 e^{kt}\right)$$

$$= y_0 k e^{kt} = ky$$

Example 3.3

The velocities at a point in a fluid in the Eulerian system are given by

$$u = x + y + z + t, \quad v = 2(x + y + z) + t, \; w = 3\,(x + y + z) + t$$

Show that the displacements of a fluid particle in the Lagrangian system are

$$x = \frac{5}{6}x_0 - \frac{1}{6}y_0 - \frac{1}{6}z_0 + \frac{1}{6}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{12}t + \frac{1}{4}t^2 - \frac{1}{72}$$

$$y = -\frac{1}{3}x_0 + \frac{2}{3}y_0 - \frac{1}{3}z_0 + \frac{1}{3}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{6}t - \frac{1}{36}$$

$$z = -\frac{1}{2}x_0 - \frac{1}{2}y_0 + \frac{1}{2}z_0 + \frac{1}{2}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{4}t + \frac{1}{4}t^2 - \frac{1}{24}$$

where x_0, y_0, and z_0 are the initial position in the Lagrangian coordinates when $t = t_0 = 0$.

Solution

$$u + v + w = 6(x + y + z) + 3t$$

or
$$\frac{dx}{dt} + \frac{dy}{dt} + \frac{dz}{dt} = 6(x + y + z) + 3t$$

or
$$\frac{d}{dt}(x + y + z) = 6(x + y + z) + 3t$$

$$x + y + z = c\,e^{6t} + \frac{3}{36}(-6t - 1) = ce^{6t} - \frac{1}{12}(6t + 1)$$

or
$$x + y + z = ce^{6t} - \frac{1}{12}(6t + 1)$$

At $t = t_0$, $x = x_0$, $y = y_0$, and $z = z_0$, which implies

$$x_0 + y_0 + z_0 = c - \frac{1}{12}$$

or
$$c = x_0 + y_0 + z_0 + \frac{1}{12}$$

Thus,
$$x + y + z = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{12}(6t + 1)$$

Now,

$$u = \frac{dx}{dt} = x + y + z + t$$

or
$$\frac{dx}{dt} = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{12}(6t + 1) + t$$

Integrating, we get

$$x = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{e^{6t}}{6} - \frac{1}{12}\left(6\frac{t^2}{2} + t\right) + \frac{t^2}{2} + c_1$$

The integration constant c_1 can be found from the initial conditions as $t = t_0 = 0, x = x_0$

$$x_0 = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{1}{6} + c_1$$

or $$c_1 = \frac{5}{6}x_0 - \frac{1}{6}y_0 - \frac{1}{6}z_0 - \frac{1}{72}$$

Putting the value of c_1, we obtain

$$x = \frac{5}{6}x_0 - \frac{1}{6}y_0 - \frac{1}{6}z_0 + \frac{1}{6}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{12}t + \frac{1}{4}t^2 - \frac{1}{72}$$

Again $$v = \frac{dy}{dt} = 2(x + y + z) + t$$

or $$\frac{dy}{dt} = 2\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{6}(6t + 1) + t$$

Integrating, we obtain

$$v = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{e^{6t}}{3} - \frac{1}{6}\left(6\frac{t^2}{2} + t\right) + \frac{t^2}{2} + c_2$$

The integration constant c_2 can be found from the initial conditions as $t = t_0 = 0, y = y_0$

$$y_0 = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{1}{3} + c_2$$

or $$c_2 = -\frac{1}{3}x_0 + \frac{2}{3}y_0 - \frac{1}{3}z_0 - \frac{1}{36}$$

Putting the value of c_2, we obtain

$$y = -\frac{1}{3}x_0 + \frac{2}{3}y_0 - \frac{1}{3}z_0 + \frac{1}{3}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{6}t - \frac{1}{36}$$

Again $$w = \frac{dz}{dt} = 3(x + y + z) + t$$

or $$\frac{dz}{dt} = 3\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{4}(6t + 1) + t$$

Integrating, we obtain

$$z = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{e^{6t}}{2} - \frac{1}{4}\left(6\frac{t^2}{2} + t\right) + \frac{t^2}{2} + c_3$$

The integration constant c_3 can be found from the initial conditions as $t = t_0, z = z_0$

$$z_0 = \left(x_0 + y_0 + z_0 + \frac{1}{12}\right)\frac{1}{2} + c_3$$

or $$c_3 = -\frac{1}{2}x_0 - \frac{1}{2}y_0 + \frac{1}{2}z_0 - \frac{1}{24}$$

Putting the value of c_3, we obtain

$$z = -\frac{1}{2}x_0 - \frac{1}{2}y_0 + \frac{1}{2}z_0 + \frac{1}{2}\left(x_0 + y_0 + z_0 + \frac{1}{12}\right)e^{6t} - \frac{1}{4}t + \frac{1}{4}t^2 - \frac{1}{24}$$

3.3.1 Variation of Flow Parameters in Time and Space

In general, the flow velocity and other hydrodynamic parameters like pressure and density may vary from one point to another at any instant, and also from one instant to another at a fixed point. According to the type of variations, different categories of flow are described as follows:

3.3.1.1 *Steady and Unsteady Flows*

A steady flow is defined as a flow in which the flow velocities and fluid properties at any point do not change with time. The flow in which any of these parameters change with time is termed as unsteady flow. In the Eulerian approach, a steady flow is described as

$$\vec{V} = \vec{V}\left(\vec{S}\right)$$

which means that the flow velocity is a function of space coordinates only. This implies that, in a steady flow, the flow velocity and fluid properties may vary with location, but the spatial distribution of any such parameter essentially remains invariant with time.

In the Lagrangian approach, time is inherent in describing the trajectory of any particle (Eq. (3.1)). But in steady flow, the velocities of all particles passing through any fixed point at different times will be same. In other words, the description of velocity as a function of time for a given particle will simply show the velocities at different points through which the particle has passed and thus furnish the information of velocity as a function of spatial location as described by the Eulerian method. Therefore, the Eulerian and the Lagrangian approaches of describing fluid motion become identical under this situation.

In practice, absolute steady flow is the exception rather than the rule, but many problems may be studied effectively by assuming that the flow is steady. Though minor fluctuations of velocity and other quantities with time occur in reality, the average value of any quantity over a reasonable interval of time remains unchanged. Moreover, a particular flow may appear steady to one observer but unsteady to another. This is because all movement is relative. The motion of a body or a fluid element is described with respect to a reference frame. Therefore, the flow may appear unsteady depending upon the choice of the reference frame. For example, the movement of water past the sides of a motorboat traveling at constant velocity would (apart from small fluctuations) appear steady to an observer in the boat. He/she would compare the water flow with an imaginary set of reference frame fixed to the boat. To an observer on a bridge, however, the same flow would appear to change with time as the boat passes underneath him/her. He/she would be comparing the flow with reference axes fixed to the bridge. Since the examination of steady flow is usually much simpler that that of unsteady flow, reference frames are chosen, where possible, so that flow with respect to the reference frame becomes steady .

3.3.1.2 *Uniform and Non-uniform Flow*

When velocity and fluid properties at any instant of time, do not change from point to point in flow field, the flow is said to be uniform. If, however, changes do occur from one point to another, the flow is non-uniform. Hence, for a uniform flow, the velocity is a function of time only, which can be expressed in the Eulerian description as

$$\vec{V} = \vec{V}(t)$$

This implies that for a uniform flow, there will be no spatial distribution of fluid velocities and flow properties. Any such parameter will have a unique value in the entire field, which of course, may change with time if the flow is unsteady.

For non-uniform flow, the changes with position may be found either in the direction of flow or in directions perpendicular to it. The latter kind of non-uniformity is always encountered near solid boundaries past which the fluids flow. This is because all fluids possess viscosity which reduces the relative velocity (velocity relative to solid boundary) to zero at a solid boundary (no-slip condition as described in Chapter 1).

For a steady and uniform flow, velocity is neither a function of time nor of space coordinates, and hence it assumes a constant value throughout the region of flow at all times. Steadiness of flow and uniformity of flow do not necessarily go together. Any of the four combinations as shown in Table 3.1 are possible.

Table 3.1

Type	*Example*
1. Steady uniform flow	Flow at constant rate through a duct of uniform cross section. (The region close to the walls of the duct is however disregarded.)
2. Steady non-uniform flow	Flow at constant rate through a duct of non-uniform cross section (tapering pipe.)
3. Unsteady uniform flow	Flow at varying rates through a long straight pipe of uniform cross section. (Again the region close to the walls is ignored.)
4. Unsteady non-uniform flow	Flow at varying rates through a duct flow of non-uniform cross section.

3.3.2 Material Derivative and Acceleration

Let the position of a particle at any instant t in a flow field be given by the space coordinates (x, y, z) with respect to a rectangular Cartesian frame of reference. The velocity components u, v, w of the particle along the x, y and z directions, respectively, can then be written in the Eulerian form as

$$u = u(x, y, z, t)$$
$$v = v(x, y, z, t)$$
$$w = w(x, y, z, t)$$

At an infinitesimal time interval Δt later, let the particle move to a new position given by the coordinates $(x + \Delta x, y + \Delta y, z + \Delta z)$, and its velocity components at this new position be $u + \Delta u$, $v + \Delta v$ and $w + \Delta w$. Therefore, we can write

$$u + \Delta u = u(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t) \tag{3.6a}$$

$$v + \Delta v = v(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t) \tag{3.6b}$$

$$w + \Delta w = w(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t) \tag{3.6c}$$

The expansion of the right-hand side of the Eqs (3.6a) to (3.6c) in the form of Taylor's series gives

$$u + \Delta u = u(x, y, z, t) + \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \frac{\partial u}{\partial z}\Delta z + \frac{\partial u}{\partial t}\Delta t + \text{higher order terms in } \Delta x, \Delta y, \Delta z \text{ and } \Delta t \tag{3.7a}$$

$$v + \Delta v = v(x, y, z, t) + \frac{\partial v}{\partial x}\Delta x + \frac{\partial v}{\partial y}\Delta y + \frac{\partial v}{\partial z}\Delta z + \frac{\partial v}{\partial t}\Delta t + \text{higher order terms in } \Delta x, \Delta y, \Delta z \text{ and } \Delta t \tag{3.7b}$$

$$w + \Delta w = w(x, y, z, t) + \frac{\partial w}{\partial x}\Delta x + \frac{\partial w}{\partial y}\Delta y + \frac{\partial w}{\partial z}\Delta z + \frac{\partial w}{\partial t}\Delta t + \text{higher order terms in } \Delta x, \Delta y, \Delta z \text{ and } \Delta t \tag{3.7c}$$

The increment in space coordinates can be written as

$$\Delta x = u\,\Delta t, \qquad \Delta y = v\,\Delta t, \quad \Delta z = w\,\Delta t$$

Substituting the values of Δx, Δy and Δz in Eqs (3.7a) to (3.7c), we have

$$\frac{\Delta u}{\Delta t} = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} + \text{terms containing } \Delta t \text{ and its higher orders}$$

$$\frac{\Delta v}{\Delta t} = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} + \text{terms containing } \Delta t \text{ and its higher orders}$$

$$\frac{\Delta w}{\Delta t} = u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} + \frac{\partial w}{\partial t} + \text{terms containing } \Delta t \text{ and its higher orders}$$

The limiting forms of the equations as $\Delta t \to 0$ become

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \tag{3.8a}$$

$$\frac{Dv}{Dt} = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} \tag{3.8b}$$

$$\frac{Dw}{Dt} = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \tag{3.8c}$$

$$\left[\text{Since } \lim_{\Delta t \to 0}\frac{\Delta u}{\Delta t} = \frac{Du}{Dt}, \ \lim_{\Delta t \to 0}\frac{\Delta v}{\Delta t} = \frac{Dv}{Dt}, \ \lim_{\Delta t \to 0}\frac{\Delta w}{\Delta t} = \frac{Dw}{Dt},\right.$$

$$\lim_{\Delta t \to 0} \text{(terms containing } \Delta t \text{ and its higher orders)} = 0 \Bigg]$$

It is important to note in this context that in the limit as $\Delta t \to 0$, the Eulerian and the Lagrangian description merge (i.e., the velocity of flow at a point at any instant is the velocity of a fluid particle that exists at that point at that instant), so that $\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = u$, which is the x component of velocity at the point of concern, and similarly for the components v and w.

It is evident from the above equations that the operator for total differential with respect to time, D/Dt in a convective field is related to the partial differential $\partial/\partial t$ as

$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} \tag{3.9}$$

The total differential D/Dt is known as the *material* or *substantial derivative* with respect to time. The first term $\partial/\partial t$ on the right-hand side of Eq. (3.9) is known as temporal or local derivative which expresses the rate of change with time, at a fixed position. The last three terms on the right-hand side of Eq. (3.9) are together known as *convective derivative* which represent the time rate of change due to change in position in the field. Therefore, the terms on the left-hand sides of Eqs (3.8a) to (3.8c) are defined the as the x, y and z components of substantial or material acceleration. The first terms on the right-hand sides of Eqs (3.8a) to (3.8c) represent the respective local or temporal accelerations, while the other terms are convective accelerations. Thus we can write,

$$a_x = \frac{\mathrm{D}u}{\mathrm{D}t} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \tag{3.9a}$$

$$a_y = \frac{\mathrm{D}v}{\mathrm{D}t} = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} \tag{3.9b}$$

$$a_z = \frac{\mathrm{D}w}{\mathrm{D}t} = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \tag{3.9c}$$

(Material or substantial acceleration) = (Temporal or local acceleration) + (Convective acceleration)

In a steady flow, the temporal acceleration is zero, since the velocity at any point is invariant with time. In a uniform flow, on the other hand, the convective acceleration is zero, since the velocity components are not the functions of space coordinates. In a steady and uniform flow, both the temporal and convective acceleration vanish and hence there exists no material acceleration. Existence of the components of acceleration for different types of flow, as described in Table 3.1, is shown in Table 3.2.

Table 3.2

Type of flow	*Material acceleration*	
	Temporal	*Convective*
Steady and uniform	0	0
Steady and non-uniform	0	Exists
Unsteady and uniform	Exists	0
Unsteady and non-uniform	Exists	Exists

3.3.2.1 *Components of Acceleration in Other Coordinate Systems*

In a cylindrical polar coordinate system (Fig. 3.1 (a), the components of acceleration in r, θ and z directions can be written as

$$a_r = \frac{\mathrm{D}v_r}{\mathrm{D}t} - \frac{v_\theta^2}{r} = \frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \tag{3.10a}$$

$$a_\theta = \frac{\mathrm{D}v_\theta}{\mathrm{D}t} + \frac{v_r v_\theta}{r} = \frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \tag{3.10b}$$

$$a_z = \frac{\mathrm{D}v_z}{\mathrm{D}t} = \frac{\partial v_z}{\partial t} + V_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \tag{3.10c}$$

The term $-v_\theta^2/r$ in the Eq. (3.10a) appears due to an inward radial acceleration arising from a change in the direction of v_θ (velocity component in the azimuthal direction θ) with θ as shown in Fig. 3.1 (a). This is typically known as centripetal acceleration. In a similar fashion, the term $v_r v_\theta / r$ represents a component of acceleration in azimuthal direction caused by a change in the direction of v_r with θ (Fig. 3.1(a).

The acceleration components in a spherical polar coordinate system (Fig. 3.1(b) can be expressed as

$$a_R = \frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial R} + \frac{v_\phi}{R}\frac{\partial v_r}{\partial \phi} + \frac{v_\theta}{R \sin\phi}\frac{\partial v_r}{\partial \theta} - \frac{v_\phi^2 + v_\theta^2}{R} \tag{3.11a}$$

$$a_\phi = \frac{\partial v_\phi}{\partial t} + v_R \frac{\partial v_\phi}{\partial R} + \frac{v_\phi}{R}\frac{\partial v_\phi}{\partial_\phi} + \frac{v_\theta}{R \sin\phi}\frac{\partial v_\phi}{\partial \theta} - \frac{v_R v_\phi}{R} - \frac{v_\theta^2 \cot\phi}{R} \tag{3.11b}$$

$$a_\theta = \frac{\partial v_\theta}{\partial t} + v_R \frac{\partial v_\theta}{\partial R} + \frac{v_\phi}{R}\frac{\partial v_\theta}{\partial \phi} + \frac{v_\theta}{R \sin\phi}\frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta v_R}{R} + \frac{v_\theta v_\phi \cot\phi}{R} \tag{3.11c}$$

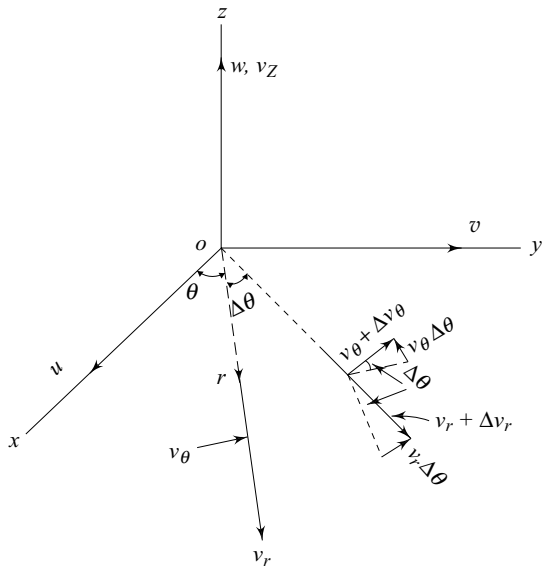


Fig. 3.1(a) Velocity components in a cylindrical polar coordinate system

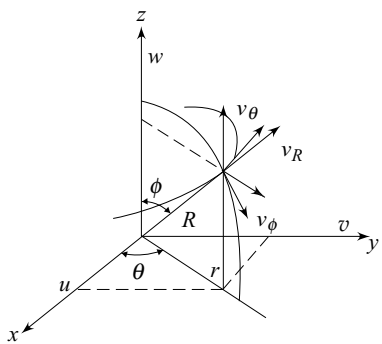


Fig. 3.1(b) Velocity components in a spherical polar coordinate system

Example 3.4

Given a velocity field $\vec{V} = (4 + xy + 2t)\,\hat{i} + 6x^3\,\hat{j} + (2xt^2 + z)\,\hat{k}$. Find the acceleration of a fluid particle at (2, 4, – 4) and time $t = 3$.

Solution

$$\vec{a} = \frac{\mathrm{D}\vec{V}}{\mathrm{D}t} = \frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla)\vec{V}$$

$$\frac{\partial \vec{V}}{\partial t} = 2\hat{i} + 6x\,t\hat{k}$$

$$(\vec{V}.\nabla)\vec{V} = \left(u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}\right)[(4 + xy + 2t)\hat{i} + 6x^3\hat{j} + (2xt^2 + z)\hat{k}]$$

$$= u(y\hat{i} + 18x^2\hat{j} + 3t^2\hat{k}) + v(x\hat{i}) + w(\hat{k})$$

$$= (4 + xy + 2t)\ (y\hat{i} + 18x^2\hat{j} + 3t^2\hat{k}) + 6x^3(x\hat{i}) + (2xt^2 + z)\hat{k}$$

$$= (4y + xy^2 + 2ty + 6x^4)\ \hat{i} + (72x^2 + 18x^3y + 36tx^2)\ \hat{j}$$

$$+ (12t^2 + 3xyt^2 + 6t^3 + 3xt^2 + z)\,\hat{k}$$

Finally, the acceleration field can be expressed as

$$\vec{a} = (2 + 4y + xy^2 + 2ty + 6x^4)\,\hat{i} + (72x^2 + 18x^3y + 36tx^2)\,\hat{j} + (6xt + 12t^2 + 3xyt^2 + 6t^3 + 3xt^2 + z)\,\hat{k}$$

The acceleration vector at the point (2, 4, –4) and at time $t = 3$ can be found out by substituting the values of x, y, z and t in the Eq. (3.36) as

$$\vec{a} = 170\,\hat{i} + 1296\,\hat{j} + 572\,\hat{k}$$

Hence, x component of acceleration $a_x = 170$ units
y component of acceleration $a_y = 1296$ units
z component of acceleration $a_z = 572$ units
Magnitude of resultant acceleration

$$|\vec{a}| = [(170)^2 + (1296)^2 + (572)^2]^{1/2}$$

$$= 1375.39 \text{ units}$$

Example 3.5

Find the acceleration components at a point (1, 1, 1) for the following flow field:

$$u = 2x^2 + 3y, \quad v = -2xy + 3y^2 + 3zy, \quad w = -\frac{3}{2}z^2 + 2xz - 9y^2z$$

Solution

x component of acceleration

$$a_x = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$$

$$= 0 + (2x^2 + 3y)\,4x + (-2xy + 3y^2 + 3zy)3 + 0$$

Therefore, $(a_x)_{\text{at }(1,\,1,\,1)} = 0 + 5 \times 4 + 4 \times 3 + 0 = 32$ units

y component of acceleration

$$a_y = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}$$

$$= 0 + (2x^2 + 3y)\,(-2y) + (-2xy + 3y^2 + 3zy)\,(-2x + 6y + 3z)$$

$$+\left(-\frac{3}{2}z^2+2xz-9y^2z\right)3y$$

Therefore, $(a_y)_{\text{at }(1,\,1,\,1)} = 5\times(-2)+4\times 7+(-8.5)\times 3 = -7.5$ units

z component of acceleration

$$\begin{aligned} a_z &= \frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+\mathrm{v}\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z} \\ &= 0+(2x^2+3y)\,2z+(-2xy+3y^2+3zy)\,(-18yz) \\ &\quad +\left(-\frac{3}{2}z^2+2xz-9y^2z\right)(-3z+2x-9y^2) \end{aligned}$$

Therefore

$$\begin{aligned} (a_z)_{\text{ at }(1,\,1,\,1)} &= 0+(2+3)\times 2-(-2+3+3)18 \\ &\quad +\left(-\frac{3}{2}+2-9\right)(-3+2-9) \\ &= 23 \text{ units} \end{aligned}$$

Example 3.6

The velocity and density fields in a diffuser are given by

$$u = u_0 e^{-2x/L} \qquad \text{and} \qquad \rho = \rho_0 e^{-x/L}$$

Find the rate of change of density at $x = L$.

Solution

The rate of change of density in this case can be written as

$$\begin{aligned} \frac{\mathrm{D}\rho}{\mathrm{D}t} &= \frac{\partial\rho}{\partial t}+u\frac{\partial\rho}{\partial x} \\ &= 0+u_0 e^{-2x/L}\,\frac{\partial}{\partial x}\left(\rho_0 e^{-x/L}\right) \\ &= u_0 e^{-2x/L}\left(-\frac{\rho_0}{L}\right)e^{-x/L} \\ &= -\frac{\rho_0 u_0}{L}e^{-3x/L}a \end{aligned}$$

$$\text{at } x = L, \quad \mathrm{D}\rho/\mathrm{D}t = -\frac{\rho_0 u_0}{L}e^{-3}$$

Example 3.7

Find the acceleration of a fluid particle at the point $r = 2a$, $\theta = \pi/2$ for a two-dimensional flow given by

$$v_r = -u\left(1-\frac{a^2}{r^2}\right)\cos\theta\,, \quad v_\theta = u\left(1+\frac{a^2}{r^2}\right)\sin\theta$$

Solution

$$\frac{\partial v_r}{\partial \theta} = u \sin\theta\left(1-\frac{a^2}{r^2}\right), \quad \frac{\partial v_\theta}{\partial \theta} = u\cos\theta\left(1+\frac{a^2}{r^2}\right)$$

$$\frac{\partial v_r}{\partial r} = \frac{-2ua^2}{r^3}\cos\theta, \quad \frac{\partial v_\theta}{\partial r} = \frac{-2ua^2}{r^3}\sin\theta,$$

Acceleration in the radial direction

$$a_r = v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r}$$

$$= \frac{2u^2a^2}{r^3}\left(1-\frac{a^2}{r^2}\right)\cos^2\theta + \frac{u^2}{r}\left(1-\frac{a^4}{r^4}\right)\sin^2\theta - \frac{u^2}{r}\left(1+\frac{a^2}{r^2}\right)^2\sin^2\theta$$

Hence, (a_r) at $r = 2a$, $\theta = \pi/2$

$$= 0 + \frac{u^2}{2a}\left(1-\frac{1}{16}\right) - \frac{u^2}{2a}\left(1+\frac{1}{4}\right)^2$$

$$= -\frac{5u^2}{16a}$$

Acceleration in the azimuthal direction

$$a_\theta = v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r}$$

$$= \frac{2u^2a^2}{r^3}\left(1-\frac{a^2}{r^2}\right)\sin\theta\cos\theta + \frac{u^2}{r}\left(1+\frac{a^2}{r^2}\right)^2\sin\theta\cos\theta$$

$$- \frac{u^2}{r}\left(1-\frac{a^4}{r^4}\right)\sin\theta\cos\theta$$

Hence, $(a_\theta)_{(\text{at } r = 2a,\ \theta = \pi/2)} = 0 + 0 + 0 = 0$

Therefore, at $r = 2a$, $\theta = \pi/2$

$$a_r = -5u^2/16, \qquad a_\theta = 0$$

Example 3.8

A fluid is flowing at a constant volume flow rate of Q through a divergent pipe having inlet and outlet diameters of D_1 and D_2, respectively, and a length of L. Assuming the velocity to be axial and uniform at any section, show that the accelerations at the inlet and outlet of the pipe are given by $-\frac{32Q^2(D_2 - D_1)}{\pi^2 L D_1^5}$ and $-\frac{32Q^2(D_2 - D_1)}{\pi^2 L D_2^5}$, respectively.

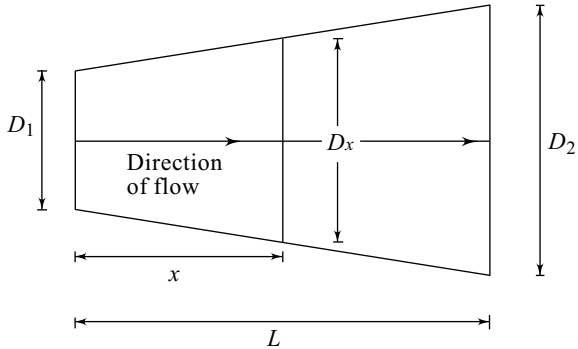


Fig. 3.2 Flow through a divergent duct

Solution

The diameter of the duct at an axial distance x from the inlet plane (Fig. 3.2) is given by

$$D_x = D_1 + \frac{x}{L}(D_2 - D_1)$$

Therefore, the velocity at this section can be written as

$$u = \frac{4Q}{\pi\left[D_1 + \frac{x}{L}(D_2 - D_1)\right]^2}$$

Acceleration at this section can be written as

$$a = u\,\frac{\partial u}{\partial x}$$

$$= \frac{4Q}{\pi\left[D_1 + \frac{x}{L}(D_2 - D_1)\right]^2} \times \frac{-8Q}{\pi\left[D_1 + \frac{x}{L}(D_2 - D_1)\right]^3} \frac{(D_2 - D_1)}{L}$$

$$= \frac{-32\,Q^2(D_2 - D_1)}{\pi^2 L\left[D_1 + \frac{x}{L}(D_2 - D_1)\right]^5}$$

This is the general expression of acceleration at any section at a distance x from the inlet of the pipe. Substituting the values of $x = 0$ (for inlet) and $x = L$ (for outlet) in the above equation, we have

$$\text{Acceleration at the inlet} = \frac{-32\,Q^2(D_2 - D_1)}{\pi^2 L D_1^5}$$

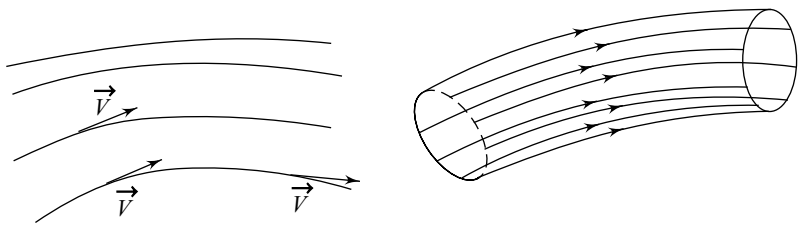
$$\text{and acceleration at the outlet} = \frac{-32\,Q^2(D_2 - D_1)}{\pi^2 L D_2^5}$$

3.3.3 Streamlines, Path Lines and Streak Lines

3.3.3.1 *Streamlines*

The analytical description of flow velocities by the Eulerian approach is geometrically depicted through the concept of streamlines. In the Eulerian method, the velocity vector is defined as a function of time and space coordinates. *If for a fixed instant of time, a space curve is drawn so that it is tangent everywhere to the velocity vector, then this curve is called a streamline. Therefore, the Eulerian method gives a series of instantaneous streamlines of the state of motion (Fig. 3.3(a)). In other words, a streamline at any instant can be defined as an imaginary curve or line in the flow field so that the tangent to the curve at any point represents the direction of the instantaneous velocity at that point.* In an unsteady flow where the velocity vector changes with time, the pattern of streamlines also changes from instant to instant. In a steady flow, the orientation or the pattern of streamlines will be fixed. From the above definition of streamline, it can be written

$$\vec{V} \times \mathrm{d}\vec{S} = 0 \tag{3.12}$$



(a) Streamlines (b) Stream tube

Fig. 3.3

$\mathrm{d}\vec{S}$ is the length of an infinitesimal line segment along a streamline at a point where $\vec{V}$ is the instantaneous velocity vector. The above expression therefore represents the differential equation of a streamline. In a Cartesian coordinate system, the vectors $\vec{V}$ and $\mathrm{d}\vec{S}$ can be written in terms of their components along the coordinate axes as $\vec{V} = \hat{i}u + \hat{j}v + \hat{k}w$ and $\mathrm{d}\vec{S} = \hat{i}\,\mathrm{d}x + \hat{j}\,\mathrm{d}y + \hat{k}\,\mathrm{d}z$. Then Eq. (3.12) gives

$$\frac{\mathrm{d}x}{u} = \frac{\mathrm{d}y}{v} = \frac{\mathrm{d}z}{w} \tag{3.13}$$

and thus describes the differential equation of streamlines in a cartesian frame of reference.

A bundle of neighbouring streamlines may be imagined to form a passage through which the fluid flows (Fig. 3.3(b)). This passage (not necessarily circular in cross section) is known as a stream tube. A stream tube with a cross section small enough for the variation of velocity over it to be negligible is sometimes termed as a stream filament. Since the stream tube is bounded on all sides by streamlines, and

by definition, velocity does not exist across a streamline, no fluid may enter or leave a stream tube except through its ends. The entire flow in a flow field may be imagined to be composed of flows through stream tubes arranged in some arbitrary positions.

3.3.3.2 *Path Lines*

Path lines are the outcome of the Lagrangian method in describing fluid flow and show the paths of different fluid particles as a function of time. In other words, *a path line is the trajectory of a fluid particle of fixed identity as defined by Eq. (3.1). Therefore a family of path lines represents the trajectories of different particles, say, P_1, P_2, P_3, etc. (Fig. 3.4)*. It can be mentioned in this context that while stream lines are referred to a particular instant of time, the description of path lines *inherently* involves *the variation of time*, since a fluid particle takes time to move from one point to another. Two path lines can intersect with one another or *a single path line itself can form a loop*. This is quite possible in a sense that, under certain conditions of flow, different particles *or even a same particle can arrive at same location at different instants of time*. The two stream lines, on the other hand, can never intersect each other since *the instantaneous velocity vector at a given location is always unique*. It is evident that path lines are identical to streamlines *in a steady flow as the Eulerian and the Lagrangian versions become the same.*

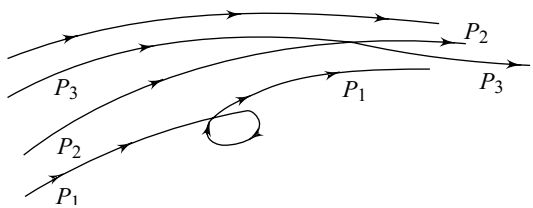


Fig. 3.4 Path lines

3.3.3.3 *Streak Lines*

A streak line at any instant of time is the locus of the temporary locations of all particles that have passed though a fixed point in the flow field. While a path line refers to the identity of a fluid particle, a streak line is specified by a fixed point in the flow field. This line is of particular interest in experimental flow visualisation. If dye is injected into a liquid at a fixed point in the flow field, then at a later time t, the dye will indicate the end points of the path lines of particles which have passed through the injection point. The equation of a streak line at time t can be derived by the Lagrangian method. If a fluid particle ($\vec{S}_0$) passes through a fixed point ($\vec{S}_1$) in a course of time t, then the Lagrangian method of description gives the equation

$$S(\vec{S}_0, t) = \vec{S}_1 \tag{3.14}$$

or solving for $\vec{S}_0$,

$$\vec{S}_0 = F(\vec{S}_1, t) \tag{3.15}$$

If the positions ($\vec{S}$) of the particles which have passed through the fixed point ($\vec{S}_1$) are determined, then a streak line can be drawn through these points. The equation of the streak line at a time t is given by

$$\vec{S} = f(\vec{S}_0, t) \tag{3.16}$$

Upon substitution of Eq. (3.15) into Eq. (3.16) we obtain,

$$\vec{S} = f[F(\vec{S}_1, t), t] \tag{3.17}$$

This is the final form of the equation of a streak line referred to a fixed point S_1. Figure 3.5 describe the difference between streak lines and path lines. Let P be a fixed point in space through which particles of different identities pass at different times. In an unsteady flow, the velocity vector at P will change with time and hence the particles arriving at P at different times will traverse different paths like PAQ, PBR, and PCS which represent the path lines of the particles. Let at any instant t_1, these particles arrive at points Q, R, and S. Thus, Q, R, and S represent the end points of the trajectories of these three particles at the instant t_1. Therefore, the curve joining the points S, R, Q, and the fixed point P will define the streak line at that instant t_1. The fixed point P will also lie on the line, since at any instant, there will be always a particle of some identity at that point. For a steady flow, the velocity vector at any point is invariant with time and hence the path lines of the particles with different identities passing through P at different times will not differ, rather would coincide with one another in a single curve which will indicate the streak line too. Therefore, in a steady flow, the path lines, streak lines and streamlines are identical.

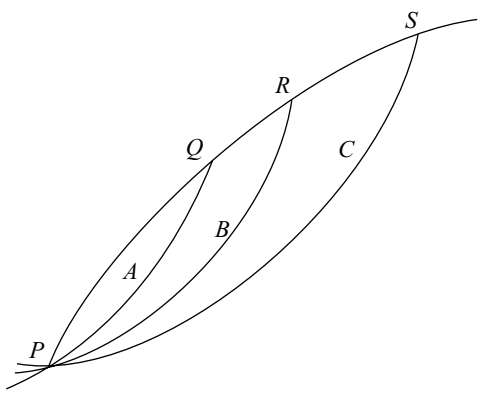


Fig. 3.5 Description of a streakline

Example 3.9

A two-dimensional flow field is defined as $\vec{V} = \hat{i}x - \hat{j}y$. Derive the equation of stream line passing through the point (1, 1).

Solution

The equation of stream line is

$$\vec{V} \times d\vec{s} = 0$$

or, $$udy - vdx = 0$$

Hence, $$dy/dx = v/u = -y/x$$

or, $$dy/y + dx/x = 0$$

Integration of the above equation gives $xy = C$, where C is constant

For the stream line passing through (1,1), the value of the constant C is 1. Hence, the required equation of stream line passing through (1,1) is $xy - 1 = 0$.

Example 3.10

The velocity components in a flow field are given as follows:

$u = \dfrac{x}{t + t_0}$, $v = v_0$, $w = 0$, where t_0 and v_0 are constants. A coloured dye is injected at the point A (x_0, y_0, z_0) in the flow field. Find (i) the equation of a coloured line visible in the flow field at $t = 2t_0$, as a consequence of the dye injection, given that the dye injection starts at $t = t_0$; and (ii) Find the locus of a fluid particle that passes through the point A at $t = t_0$.

Solution

(i) Here, essentially, we need to obtain the equation of streakline at $t = 2t_0$. In order to obtain the same, we consider that fluid particles are injected through the point at instants of time symbolically denoted by t_i, where $t_i \leq 2t_0$. Importantly t_i is a variable and not a fixed instant of time. Now, based on the given velocity field, we have

$$u = \frac{dx}{dt} = \frac{x}{t + t_0}$$

$$\Rightarrow \quad \int_{x_0}^{x} \frac{dx}{x} = \int_{t_i}^{2t_0} \frac{dt}{t + t_0}$$

$$\Rightarrow \quad \ln \frac{x}{x_0} = \ln \frac{2t_0 + t_0}{t_i + t_0}$$

$$\Rightarrow \quad \frac{x}{x_0} = \frac{3t_0}{t_i + t_0} \tag{3.18}$$

Similarly, $$v = \frac{dy}{dt} = v_0$$

$$\Rightarrow \quad \int_{y_0}^{y} dy = \int_{t_i}^{2t_0} v_0 dt$$

$$\Rightarrow \qquad y - y_0 = v_0 (2t_0 - t_i)$$

$$\Rightarrow \qquad t_i = 2t_0 + \frac{y - y_0}{v_0} \tag{3.19}$$

Eliminating t_i from Eqs (3.18) and (3.19), one can write

$$\frac{x}{x_0} = \frac{3t_0}{2t_0 + \dfrac{y - y_0}{v_0} + t_0} = \frac{3t_0}{3t_0 + \dfrac{y - y_0}{v_0}} = \frac{1}{1 + \dfrac{y - y_0}{3t_0 v_0}}$$

$$\frac{x}{x_0}\left(1 + \frac{y - y_0}{3t_0 v_0}\right) = 1 \tag{3.20}$$

Equation. (3.20) is the equation of a coloured line visible in the flow field at $t = 2t_0$.
(ii) Here, essentially, we need to obtain the equation of the pathline, corresponding to a fluid particle that at $t = t_0$ passed through (x_0, y_0)

Now, $$u = \frac{dx}{dt} = \frac{x}{t + t_0}$$

$$\Rightarrow \qquad \int_{x_0}^{x} \frac{dx}{x} = \int_{t_0}^{t} \frac{dt}{t + t_0}$$

$$\Rightarrow \qquad \ln \frac{x}{x_0} = \ln \frac{t + t_0}{t_0 + t_0} = \ln \frac{t + t_0}{2t_0}$$

$$\Rightarrow \qquad \frac{x}{x_0} = \frac{t + t_0}{2t_0} \tag{3.21}$$

Similarly, $$v = \frac{dy}{dt} = v_0$$

$$\int_{y_0}^{y} dy = \int_{t_0}^{t} v_0 dt$$

$$y - y_0 = v_0 (t - t_0)$$

$$t - t_0 = + \frac{y - y_0}{v_0} \tag{3.22}$$

Eliminating t from Eqs (3.21) and (3.22), one can write

$$\frac{x}{x_0} = \frac{t_0 + \dfrac{y - y_0}{v_0} + t_0}{2t_0} = 1 + \frac{y - y_0}{2t_0 v_0}$$

$$\frac{x}{x_0} - \frac{y - y_0}{2t_0 v_0} = 1 \tag{3.23}$$

Equation. (3.23) is the equation of the locus of a fluid particle that passes through the point A at $t = t_0$.

3.3.4 One-, Two- and Three-Dimensional Flows

In general, fluid flow is three dimensional. This means that the flow parameters like velocity, pressure and so on vary in all the three coordinate directions. Sometimes simplification is made in the analysis of different fluid flow problems by selecting the coordinate directions so that appreciable variation of the hydrodynamic parameters take place in only two directions or even in only one.

So *one-dimensional flow* is that in which all the flow parameters may be expressed as functions of time and one space coordinate only. This single space coordinate is usually the distance measured along the centre line (not necessarily straight) of some conduit in which the fluid is flowing. For instance, the flow in a pipe is considered one-dimensional when variations of pressure and velocity occur along the length of the pipe, but any variation over the cross section is assumed negligible. In reality the flow is never one dimensional because viscosity causes the velocity to decrease to zero at the solid boundaries. If, however, the non-uniformity of the actual flow is not too great, valuable results may often be obtained from 'one-dimensional analysis'. Under this situation, the average values of the flow parameters at any given section (perpendicular to the flow) are assumed to be applied to the entire flow at that section. In two-dimensional flow, the flow parameters are functions of time and two space coordinates (say, x and y). There is no variation in the z direction, and therefore the same streamline pattern could, at any instant, be found in all planes perpendicular to the z direction. In three-dimensional flow, the hydrodynamic parameters are functions of three space coordinates and time.

3.3.5 Translation, Rate of Deformation and Rotation

The movement of a fluid element in space has three distinct features, namely, translation, rate of deformation, and rotation. A fluid motion, in general, consists of these three features simultaneously. Translation and rotation without deformation represent rigid-body displacements which do not induce any strain in the body. Figure. 3.6 shows the picture of a pure translation in absence of rotation and deformation of a fluid element in a three-dimensional flow described by a rectangular Cartesian coordinate system. In absence of deformation and rotation, there will be no change in the length of the sides or in the included angles made by the sides of the fluid element. The sides are displaced parallely. This is possible when the flow velocities u (the x component velocity), v (the y component velocity), and w (the z component velocity) are neither a function of x, y, nor of z, in other words, the flow field is totally uniform.

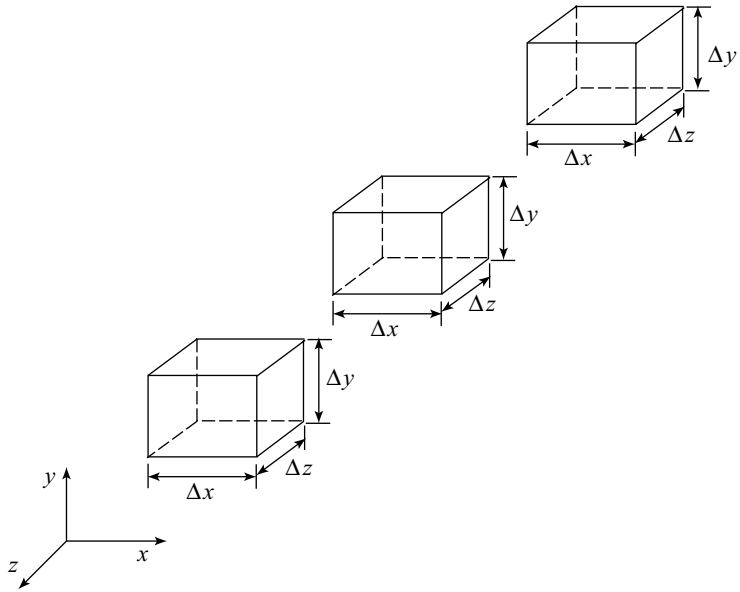


Fig. 3.6

3.3.5.1 *Rate of Linear Deformation*

Now consider a situation where a component of flow velocity becomes the function of only one space coordinate along which that velocity component is defined, for example, $u = u(x)$, $v = v(y)$ and $w = w(z)$. Let us consider a fluid element, as shown in Fig. 3.7 (a), which in course of its translation suffers a change in its linear dimensions without any change in the included angle by the sides.

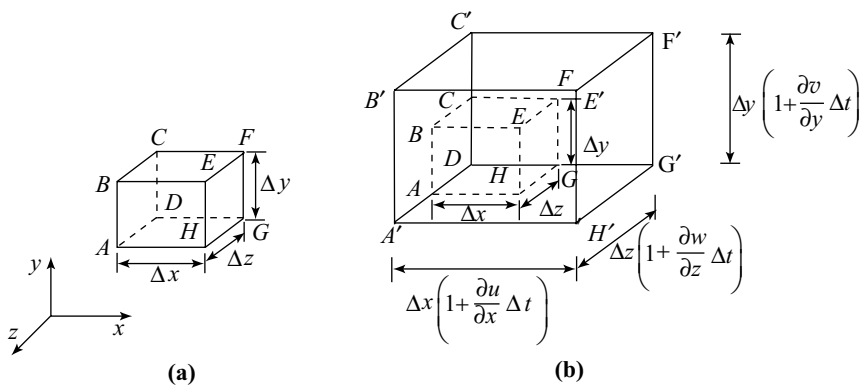


Fig. 3.7

Displacement of the faces ABCD and EFGH along the x direction during a time interval of Δt are $u\Delta t$ and $\left(u+\frac{\partial u}{\partial x}\Delta x\right)\Delta t$, respectively (neglecting higher order terms). As a result, there occurs a change in length of the element along the x direction which is equal to $\frac{\partial u}{\partial x}\Delta x\Delta t$ (Fig.3.7(b)). Therefore, the rate of elemental strain (the rate of change of length per unit original length) along the x direction is

$$\dot{\varepsilon}_x = \frac{\partial u}{\partial x} \tag{3.24a}$$

In a similar way, the rate of strain along the y and z directions can be expressed as

$$\dot{\varepsilon}_y = \frac{\partial v}{\partial y} \tag{3.24b}$$

$$\dot{\varepsilon}_z = \frac{\partial w}{\partial z} \tag{3.24c}$$

New lengths along the x, y and z directions (Fig.3.7(b)) are $\Delta x\left(1+\frac{\partial u}{\partial x}\Delta t\right)$, $\Delta y\left(1+\frac{\partial v}{\partial y}\Delta t\right)$ and $\Delta z\left(1+\frac{\partial w}{\partial z}\Delta t\right)$, respectively. Therefore, the change in volume of the fluid element (Fig. 3.7 (b)) is

$$\Delta x\Delta y\Delta z\left(1+\frac{\partial u}{\partial x}\Delta t\right)\left(1+\frac{\partial v}{\partial y}\Delta t\right)\left(1+\frac{\partial w}{\partial z}\Delta t\right)-\Delta x\Delta y\Delta z$$

$$=\Delta x\Delta y\Delta z\left[\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}\right)\Delta t\right] \text{ (neglecting higher order terms)}$$

Rate of volumetric strain is given by the rate of change of volume per unit original volume, which becomes

$$\dot{\varepsilon}_{\text{vol}} = \frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z} = \dot{\varepsilon}_x+\dot{\varepsilon}_y+\dot{\varepsilon}_z$$

If the volume of the fluid element is denoted by $\forall$, then by definition $\dot{\varepsilon}_{\text{vol}} = \frac{1}{\forall}\frac{D\forall}{Dt}$.

Thus,

$$\frac{1}{\forall}\frac{D\forall}{Dt} = \frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z} = \nabla.\vec{V} \tag{3.24d}$$

The above implicates that the volumetric strain of a fluid element is given by the divergence of the velocity vector. Flows in which there is no volumetric strain are known as incompressible flows. Incompressible flows therefore, by definition are flows with $\nabla.\vec{V}=0$. It is important to note that there is a subtle difference between incompressible fluid, and incompressible flows. Flows with zero rate of volumetric

strain are termed as incompressible flows. On contrary, a fluid is called incompressible if its density does not change significantly with change in pressure.

It is important to mention here that the condition for incompressibility is intimately related to the consideration of conservation of mass for incompressible flows. In Section 3.4 we will revisit this issue with more elaborate details, starting from the general principles of conservation of mass applied to fluid flow.

3.3.5.2 *Rate of Angular Deformation and Rotation of Fluid Elements*

For simplicity, let us consider a fluid element ABCD (Fig. 3.8) in a two-dimensional velocity field, where both the velocities u and v become functions of x and y, i.e.,

$$u = u(x, y)$$
$$v = v(x, y)$$

In the expansion of u and v about x and y in a Taylor series function, the higher order terms are neglected for any derivation as follows. Displacement of A and B along the x direction during a time duration Δt are $u\Delta t$ and $\left(u + \frac{\partial u}{\partial x}\Delta x\right)\Delta t$, respectively. Similarly, the displacement of A and B along the y direction during a time duration Δt are $v\Delta t$ and $\left(v + \frac{\partial v}{\partial x}\Delta x\right)\Delta t$, respectively. Therefore, the point B has relative displacements in both x and y directions with respect to point A, whose magnitudes are $\frac{\partial u}{\partial x}\Delta x\Delta t$ and $\frac{\partial v}{\partial x}\Delta x\Delta t$, respectively. Similarly, point D has relative displacements in both x and y directionss with respect to point A, whose magnitudes are $\frac{\partial u}{\partial y}\Delta y\Delta t$ and $\frac{\partial v}{\partial y}\Delta y\Delta t$, respectively. Because of the relative displacement of B in the y direction with respect to A and the relative displacement of D in the x direction with respect to A, the included angle between AB and AD changes, and the fluid element suffers a continuous angular deformation along with the linear deformations in course of its motion. The rate of angular strain or angular deformation $\dot{\varepsilon}_{xy}$ is defined as the rate of change of the angle between two line elements in the fluids which were originally perpendicular to each other. Consider AB and AD as two such line elements, for example. On deformation, the angle between them is, $< \mathrm{B'A'D'} = \frac{\pi}{2} - (\Delta\alpha + \Delta\beta)$.

The change in the angle is $= \frac{\pi}{2} - \left[\frac{\pi}{2} - (\Delta\alpha + \Delta\beta)\right] = (\Delta\alpha + \Delta\beta)$

Rate of change of the angle, or rate of angular deformation is

$$\dot{\varepsilon}_{xy} = \lim_{\Delta t \to 0}\left(\frac{\Delta\alpha + \Delta\beta}{\Delta t}\right)$$

or

$$\dot{\varepsilon}_{xy} = \frac{d\alpha}{dt} + \frac{d\beta}{dt}$$

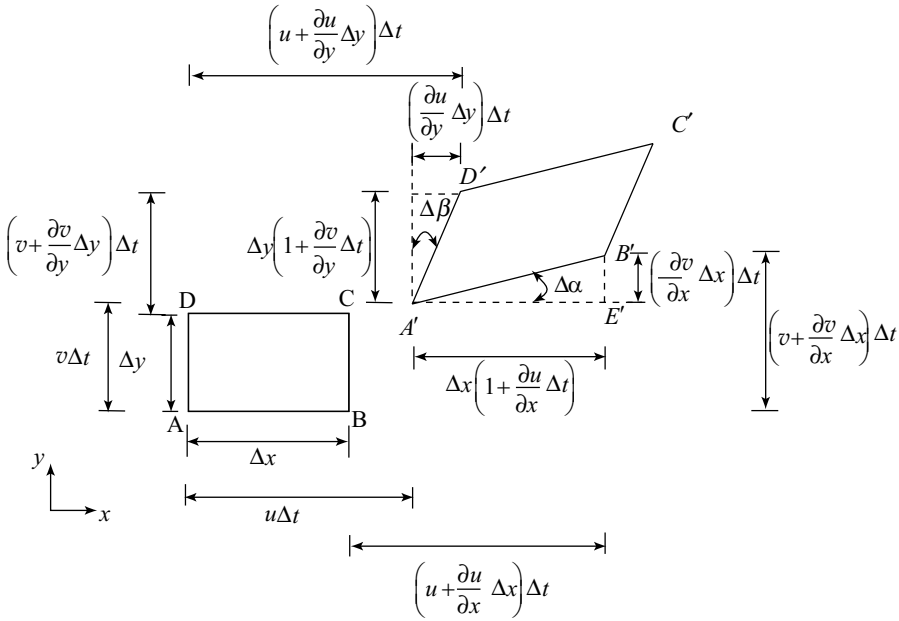


Fig. 3.8

From the geometry, we obtain

$$\tan \Delta\alpha = \frac{B'E'}{A'E'} = \frac{\frac{\partial v}{\partial x}\Delta x\Delta t}{\Delta x\left(1+\frac{\partial u}{\partial x}\Delta t\right)}$$

For small values of $\Delta\alpha$ (which occurs at small Δt), $\Delta\alpha \approx \Delta\alpha$

Hence,
$$\Delta\alpha = \frac{\frac{\partial v}{\partial x}\Delta x\Delta t}{\Delta x\left(1+\frac{\partial u}{\partial x}\Delta t\right)}$$

$$\frac{d\alpha}{dt} = \lim_{\Delta t\to 0}\frac{\Delta\alpha}{\Delta t} = \lim_{\Delta t\to 0}\frac{\frac{\partial v}{\partial x}\Delta x}{\Delta x\left(1+\frac{\partial u}{\partial x}\Delta t\right)}$$

$$= \frac{\partial v}{\partial x}$$

Similarly,
$$\frac{d\beta}{dt} = \lim_{\Delta t\to 0}\frac{\Delta\beta}{\Delta t} = \lim_{\Delta t\to 0}\frac{\frac{\partial u}{\partial y}\Delta y}{\Delta x\left(1+\frac{\partial v}{\partial y}\Delta t\right)}$$

$$= \frac{\partial u}{\partial y}$$

Thus, $$\dot{\varepsilon}_{xy} = \frac{d\alpha}{dt} + \frac{d\beta}{dt} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \tag{3.25}$$

Similarly,

$$\dot{\varepsilon}_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \tag{3.25a}$$

$$\dot{\varepsilon}_{xz} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \tag{3.25b}$$

In index notation, the rate of angular deformation can be expressed as

$$\dot{\varepsilon}_{ij} = \frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \tag{3.26}$$

Specification of $\dot{\varepsilon}$, thus requires two indices; and like stress, it is also a second order tensor. It may be easily derived that the strain rate tensor is symmetric, i.e., $\dot{\varepsilon}_{ij} = \dot{\varepsilon}_{ji}$.

It is important to note in this context that because of relative motion between fluid layers, deformable fluids do not rotate like rigid bodies. However, an angular velocity for such fluid flow may be artificially defined in such a way that in the limiting case of non-deformable fluids, it conforms to the motion of rigid body rotation. Accordingly, the angular velocity at a point is defined as the arithmetic mean of angular velocities of two line elements at that point, which were originally perpendicular to each other. Referring to Fig. 3.8, the angular velocities of two originally perpendicular line elements AB and AD in the xy plane are $\frac{d\alpha}{dt}$ and $\frac{d\beta}{dt}$ respectively, but oriented in the opposite sense. Considering the anticlockwise direction as positive, the angular velocity of the fluid element at A thus can be written as,

$$\omega_{xy} = \frac{1}{2}\left(\dot{\alpha} - \dot{\beta}\right) = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \tag{3.27}$$

Instead of ω_{xy}, one may also use an alternative notation of ω_z, considering that rotation in the xy plane and rotation with respect to the z axis is essentially the same thing.

In three-dimensional flow, the components of rotation are defined as

$$\omega_{xy} = \omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \tag{3.27a}$$

$$\omega_{yz} = \omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) \tag{3.27b}$$

$$\omega_{zx} = \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) \tag{3.27c}$$

In index notation, the angular velocity can be expressed as

$$\omega_{ij} = \frac{1}{2}\left(\frac{\partial u_j}{\partial x_i} - \frac{\partial u_i}{\partial x_j}\right) \tag{3.28}$$

Obviously the angular velocity tensor is skew-symmetric as $\omega_{ij} = -\omega_{ji}$. It is interesting to note in this context that the velocity gradient tensor $\frac{\partial u_i}{\partial x_j}$ can be decomposed into two components, viz., one symmetric and another skew-symmetric, as follows:

$$\frac{\partial u_i}{\partial x_j} = \underbrace{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)}_{\dot{\varepsilon}_{ij}} + \underbrace{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)}_{\omega_{ij}} \tag{3.29}$$

Here the symmetric component represents the rate of shear strain tensor and the skew-symmetric part represents the angular velocity tensor.

The angular velocity of fluid flow has further mathematical implications, which may be inferred by referring to the fact that following Eqs (3.27a) to (3.27c), rotation in a flow field can be expressed in a vector form as

$$\vec{\omega} = \frac{1}{2}\left(\nabla \times \vec{V}\right) \tag{3.30}$$

The quantity $\vec{\Omega} = \nabla \times \vec{V}$ is defined as the **vorticity of flow,** which is a mathematical measure of rotationalities in the flow field. In Cartesian coordinate systems

$$\vec{\Omega} = \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$= \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)\hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)\hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{k} = 2\vec{\omega} \tag{3.31}$$

If an imaginary line is drawn in the fluid so that the tangent to it at each point is in the direction of the vorticity vector $\vec{\Omega}$ at that point, the line is called a vortex line. Therefore, the general equation of the vortex line can be written as

$$\vec{\Omega} \times d\vec{s} = 0 \tag{3.32}$$

Let us now consider some special cases of angular deformation of fluid elements

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y},$$

$$\dot{\varepsilon}_{xy} = 0 \text{ (from Eq. 3.25)} \tag{3.33a}$$

and

$$\omega_z = \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \text{(from Eq. 3.27)} \tag{3.33b}$$

This implies that the linear segments $A\,B$ and $A\,D$ move with the same angular velocity (both in magnitude and direction) and hence the included angle between them remains the same and no angular deformation takes place. This situation is known as pure rotation (Fig. 3.9(a). In another special case,

when

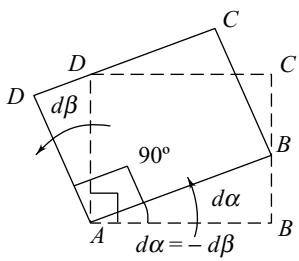
$$\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$

$$\dot{\gamma}_{xy} = 2\frac{\partial v}{\partial x} = 2\frac{\partial u}{\partial y} \quad \text{(from Eq. 3.25)} \tag{3.34a}$$

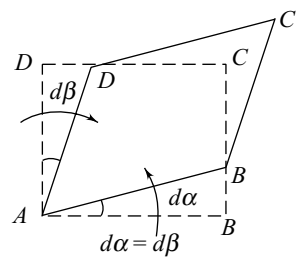
and

$$\omega_z = 0 \text{ (from Eq. (3.27))} \tag{3.34b}$$

This implies that the fluid element has an angular deformation rate but no rotation about the z axis (Fig. 3.9(b)



(a) Fluid element in pure rotation in absence of angular deformation



(b) Fluid element with angular deformation in absence of rotation

Fig. 3.9

When the components of rotation at all points in a flow field become zero, the flow is said to be irrotational. Therefore, the necessary and sufficient condition for a flow field to be irrotational is

$$\nabla \times \vec{V} = 0 \tag{3.35}$$

3.3.5.3 *Vorticity in Polar Coordinates*

In a two-dimensional polar coordinate system (Fig. 3.10), the angular velocity of segment Δr can be written as

$$\lim_{\Delta t \to 0}\left[\frac{(v_\theta + (\partial v_\theta / \partial r)\,\Delta r - v_\theta)\,\Delta t}{\Delta r\,\Delta t}\right] = \frac{\partial v_\theta}{\partial r}$$

Also, the angular velocity of segment $r\Delta\theta$ becomes

$$\lim_{\Delta t \to 0}\left[-\frac{(v_r + (\partial v_r / \partial \theta)\,\Delta\theta - v_r)\,\Delta t}{r\,\Delta\theta\,\Delta t}\right] = -\frac{1}{r}\frac{\partial v_r}{\partial \theta}$$

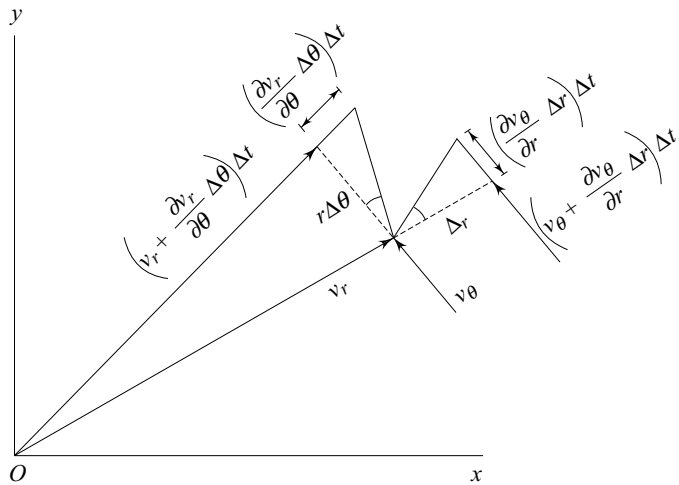


Fig. 3.10 Definition of rotation in a polar coordinate system

The additional term arising from the angular velocity about the centre O is V_θ/r.

Hence, the vorticity component Ω_z in polar coordinates is

$$\Omega_z = \; 2\omega_z = \frac{\partial v_\theta}{\partial r} - \frac{1}{r}\frac{\partial v_r}{\partial \theta} + \frac{v_\theta}{r}$$

Therefore, in a three-dimensional cylindrical polar coordinate system, the vorticity components can be written as

$$\Omega_z = \; \frac{\partial v_\theta}{\partial r} - \frac{1}{r}\frac{\partial v_r}{\partial \theta} + \frac{v_\theta}{r} \tag{3.36a}$$

$$\Omega_r = \; \frac{1}{r}\frac{\partial v_z}{\partial \theta} - \frac{\partial v_\theta}{\partial z} \tag{3.36b}$$

$$\Omega_\theta = \; \frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r} \tag{3.36c}$$

In a spherical polar coordinate system (Fig. 3.1(b)), the vorticity components are defined as

$$\Omega_R = \frac{1}{R}\frac{\partial v_\theta}{\partial \phi} - \frac{1}{R\sin\phi}\frac{\partial v_\phi}{\partial \theta} + \frac{v_\theta}{R}\cot\phi \tag{3.37a}$$

$$\Omega_\phi = \frac{1}{R\sin\phi}\frac{\partial v_R}{\partial \theta} - \frac{\partial v_\theta}{\partial R} - \frac{v_\theta}{R} \tag{3.37b}$$

$$\Omega_\theta = \frac{\partial v_\phi}{\partial R} + \frac{v_\phi}{R} - \frac{1}{R}\frac{\partial v_R}{\partial \phi} \tag{3.37c}$$

3.3.5.4 *Circulation*

The circulation Γ around a closed contour is defined as the line integral of the tangential component of velocity and is given by (Fig. 3.11)

$$\Gamma = \int \vec{V}.d\vec{s} \tag{3.38}$$

where $d\vec{s}$ is a line element of the contour. Circulation is traditionally taken to be positive in the anticlockwise direction (not a mandatory but a general convention). The circulation can be expressed in terms of area integral using Stokes theorem as

$$\Gamma = \int \vec{V}.d\vec{s} = \int_A \left(\nabla \times \vec{V}\right).\hat{n}dA \tag{3.39}$$

where dA is the elemental area of the surface and A is the total surface area bounded by the close curve around which circulation is evaluated. In Eq. (3.39), $\hat{n}$ is a unit vector outward normal to dA.

Using the definition of vorticity, Eq. (3.39) becomes

$$\Gamma = \int_A \vec{\Omega}.\hat{n}dA \tag{3.40}$$

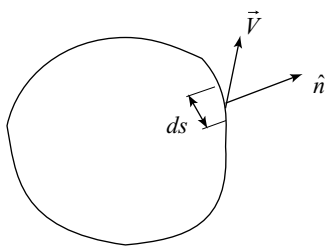


Fig. 3.11

It is interesting to illustrate the concept of circulation through some special types of flows known as vortex flows. Vortex flows are flows for which the radial component of velocity, v_r, is zero; but the cross-radial component of velocity, v_θ, is non-zero (in fact, a function of r).

The first example of vortex flow that we introduce here is called *forced vortex*, which is defined by the following velocity components:

$$v_\theta = \omega r$$

and $$v_r = 0$$

Consider a close contour $ABCD$ in the above flow field shown in Fig. 3.12. Circulation along the close contour $ABCD$ is given by

$$\Gamma_{ABCD} = \int \vec{V}.d\vec{s} = -v_{\theta_{AB}} r d\theta - v_{r_{BC}} dr + v_{\theta_{CD}} (r+dr) d\theta + v_{r_{DA}} r d\theta$$

$$= -(\omega r) r d\theta - \cancelto{0}{v_{r_{BC}}} dr + \omega (r+dr)(r+dr) d\theta - \cancelto{0}{v_{r_{DA}}} r d\theta$$

$$\Gamma_{ABCD} = 2\omega r dr d\theta \tag{3.41}$$

Circulation per unit area is given by

$$\frac{\Gamma_{ABCD}}{r dr d\theta} = 2\omega \tag{3.41a}$$

Interestingly, here 2ω is the vorticity of flow. Thus, circulation per unit area is the vorticity of the flow.

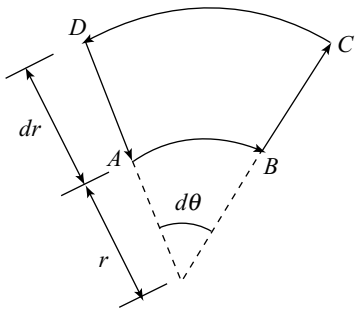


Fig. 3.12

As a next example of vortex flow, we consider the *free vortex,* the flow field corresponding to which is defined by the following velocity components:

$$v_\theta = \frac{C}{r}$$

and $$v_r = 0$$

Close to such a situation is the flow observed in a kitchen sink. For such a flow, circulation along the closed contour $ABCD$ is given by

$$\Gamma_{ABCD} = \int \vec{V}.d\vec{s} = -v_{\theta_{AB}} r d\theta - v_{r_{BC}} dr + v_{\theta_{CD}} (r+dr) d\theta + v_{r_{DA}} r d\theta$$

$$= -\frac{C}{r} r d\theta - \cancelto{0}{v_{r_{BC}}} dr + \frac{C}{r+dr}(r+dr) d\theta - \cancelto{0}{v_{r_{DA}}} r d\theta = 0$$

It is important to mention here that free vortex flow at the origin is impossible because of mathematical singularity (v_θ tends to infinity as r tends to zero). In this context, one may also note that although free vortex is an irrotational motion, the circulation around a path containing any singular point (such as the origin) is non-zero. However, that is not a technically correct way of evaluating the circulation, since the contour chosen must not include any point of singularity. Thus, if the circulation is calculated for a free vortex flow along any closed contour excluding the singular point (the origin), it should be zero, consistent with the conditions to be satisfied by an irrotational flow field.

A classical example of vortex flow in nature is a tornado. Up to the eye of a tornado, it is a forced vortex, having a strong element of rotationality. Beyond a critical radius it is like a free vortex. Thus, a tornado can be represented by a *Rankine vortex* which is a combination of forced and free vortex (Fig. 3.13). At the critical radius, v_θ evaluated from free vortex zone must be equal to that evaluated from the forced vortex zone.

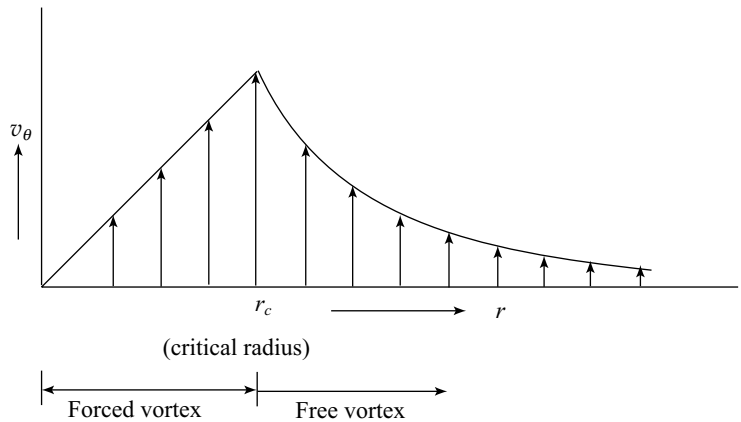


Fig. 3.13 Velocity distribution in a Rankine vortex

Example 3.11

The velocity field in a fluid medium is given by

$$\vec{V} = 3xy^2\,\hat{i} + 2xy\,\hat{j} + (2zy + 3t)\,\hat{k}$$

Find the magnitudes and directions of (i) translational velocity, (ii) rotational velocity, and (iii) the vorticity of a fluid element at (1, 2, 1) and at time $t = 3$.

Solution

(i) Translational velocity vector at (1, 2, 1) and at $t = 3$ can be written as

$$\vec{V} = 3(2)(4)\,\hat{i} + 2(1)(2)\,\hat{j} + [2(1)(2) + 3(3)]\,\hat{k} = 12\,\hat{i} + 4\,\hat{j} + 13\,\hat{k}$$

Hence, x component of translational velocity $u = 12$ units
y component of translational velocity $v = 4$ units
z component of translational velocity $w = 13$ units

(ii) Rotational velocity vector is found as

$$\vec{\omega} = \frac{1}{2}(\nabla \times \vec{V}) = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$= \frac{\hat{i}}{2}\left\{\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right\} + \frac{\hat{j}}{2}\left\{\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right\} + \frac{\hat{k}}{2}\left\{\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right\}$$

$$= \hat{i}\,\frac{1}{2}\left[\frac{\partial}{\partial y}(2zy + 3t) - \frac{\partial}{\partial z}(2xy)\right]$$

$$+ \hat{j}\,\frac{1}{2}\left[\frac{\partial}{\partial z}(3xy^2) - \frac{\partial}{\partial x}(2zy + 3t)\right]$$

$$+ \hat{k}\,\frac{1}{2}\left[\frac{\partial}{\partial x}(2xy) - \frac{\partial}{\partial y}\left(3xy^2\right)\right]$$

$$= z\,\hat{i} + (y - 3xy)\,\hat{k}$$

at (1, 2, 1) and $t = 3$,

$$\vec{\omega} = \hat{i} - 4\,\hat{k}$$

Therefore, the rotational velocity about x axis $\omega_x = 1$ unit
the rotational velocity about y axis $\omega_y = 0$ unit
the rotational velocity about z axis $\omega_z = -4$ units

(iii) The vorticity $\vec{\Omega} = 2\,\vec{\omega}$

Hence, $\vec{\Omega} = 2\,\hat{i} - 8\,\hat{k}$

Example 3.12

Find the vorticity components at a point (1,1,1) for the following flow field:

$$u = 2x^2 + 3y, \quad v = -2xy + 3y^2 + 3zy, \quad w = \frac{3}{2}z^2 + 2xz - 9y^2z$$

Solution

$$\Omega_x = \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} = 18yz - 3y = -(18yz + 3y)$$

$$\Omega_y = \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} = 0 - 2z = -2z$$

$$\Omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2y - 3 = -(2y + 3)$$

at the point (1, 1, 1)

$$\Omega_x = -(18 + 3) = -21 \text{ units}$$

$$\Omega_y = -2 \text{ units}$$

$$\Omega_z = -(2 + 3) = -5 \text{ units}$$

3.3.5.5 *Generalised Expression of the Movement of a Fluid Element*

An analytical expression to represent the most general form of the movement of a fluid element consisting of translation, rotation and deformation can be developed as follows.

Consider the movement of a fluid element in a fluid continuum as shown in Fig. 3.14.

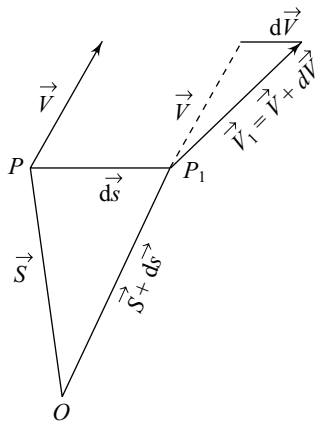


Fig. 3.14 General representation of fluid motion

The velocity at a point $P(x, y, z)$ is $\vec{V}$ and at point $P_1(x_1, y_1, z_1)$, a small distance $d\vec{S}$ from P, is $\vec{V}_1$.

The velocity vector $\vec{V}_1$ can be written as

$$\vec{V}_1 = \hat{i}u_1 + \hat{j}v_1 + \hat{k}w_1 = \vec{V} + d\vec{V}$$

$$= \hat{i}u + \hat{j}v + \hat{k}w + \frac{\partial \vec{V}}{\partial x}dx + \frac{\partial \vec{V}}{\partial y}dy + \frac{\partial \vec{V}}{\partial z}dz$$

$$= \hat{i}\left[u + \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz\right]$$

$$+\ \hat{j}\left[v+\frac{\partial v}{\partial x}\mathrm{d}x+\frac{\partial v}{\partial y}\mathrm{d}y+\frac{\partial v}{\partial z}\mathrm{d}z\right]$$
$$+\ \vec{k}\left[w+\frac{\partial w}{\partial x}\mathrm{d}x+\frac{\partial w}{\partial y}\mathrm{d}y+\frac{\partial w}{\partial z}\mathrm{d}z\right] \tag{3.42}$$

where u_1, v_1, w_1 are the respective x, y and z components of $\vec{V}_1$ and u, v, w are those of $\vec{V}$.

Equation (3.42) can be rearranged as

$$\begin{aligned}\vec{V}_1 = {} & \hat{i}\left[u+\left\{\frac{\partial u}{\partial x}\mathrm{d}x+\frac{1}{2}\left(\frac{\partial \mathrm{v}}{\partial x}+\frac{\partial u}{\partial y}\right)\mathrm{d}y+\frac{1}{2}\left(\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\right)\mathrm{d}z\right\}\right.\\
& \left.+\ \frac{1}{2}\left\{\left(\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x}\right)\mathrm{d}z-\left(\frac{\partial \mathrm{v}}{\partial x}-\frac{\partial u}{\partial y}\right)\mathrm{d}y\right\}\right]\\
& +\ \hat{j}\left[v+\left\{\frac{\partial v}{\partial y}\mathrm{d}y+\frac{1}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\mathrm{d}x+\frac{1}{2}\left(\frac{\partial w}{\partial y}+\frac{\partial v}{\partial z}\right)\mathrm{d}z\right\}\right.\\
& \left.+\ \frac{1}{2}\left\{\left(\frac{\partial \mathrm{v}}{\partial x}-\frac{\partial u}{\partial y}\right)\mathrm{d}x-\left(\frac{\partial w}{\partial y}-\frac{\partial \mathrm{v}}{\partial z}\right)\mathrm{d}z\right\}\right]\\
& +\ \hat{k}\left[w+\left\{\frac{\partial w}{\partial z}\mathrm{d}z+\frac{1}{2}\left(\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\right)\mathrm{d}x+\frac{1}{2}\left(\frac{\partial w}{\partial y}+\frac{\partial \mathrm{v}}{\partial z}\right)\mathrm{d}y\right\}\right.\\
& \left.+\ \frac{1}{2}\left\{\left(\frac{\partial w}{\partial y}-\frac{\partial \mathrm{v}}{\partial z}\right)\mathrm{d}y-\left(\frac{\partial u}{\partial z}-\frac{\partial w}{\partial x}\right)\mathrm{d}x\right\}\right]\\
= {} & \vec{V}+\frac{1}{2}\vec{\Omega}\times\mathrm{d}\vec{s}+\vec{D}\end{aligned} \tag{3.43}$$

where, $\mathrm{d}\vec{S} = \hat{i}\,\mathrm{d}x+\hat{j}\,\mathrm{d}y+\hat{k}\,\mathrm{d}z$,

$\vec{\Omega}$ is the vorticity vector as defined by Eq. (3.31)

and
$$\begin{aligned}\vec{D} = {} & \hat{i}\left[\frac{\partial u}{\partial x}\mathrm{d}x+\frac{1}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\mathrm{d}y+\frac{1}{2}\left(\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\right)\mathrm{d}z\right]\\
& +\ \hat{j}\left[\frac{1}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\mathrm{d}x+\frac{\partial v}{\partial y}\mathrm{d}y+\frac{1}{2}\left(\frac{\partial w}{\partial y}+\frac{\partial v}{\partial z}\right)\mathrm{d}z\right]\\
& +\ \hat{k}\left[\frac{1}{2}\left(\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\right)\mathrm{d}x+\frac{1}{2}\left(\frac{\partial w}{\partial y}+\frac{\partial v}{\partial z}\right)\mathrm{d}y+\frac{\partial w}{\partial z}\mathrm{d}z\right]\end{aligned} \tag{3.44}$$

Equation (3.43) represents the most general form of the movement of a fluid element. The first term represents the translational velocity which indicates linear motion without any change of shape of the fluid body. The second term represents a rigid body rotation of the fluid element, while the third term $\vec{D}$ represents the rate of deformation.

3.4 CONSERVATION OF MASS FOR FLUID FLOW

A fluid being a material body, must obey the law of conservation of mass in the course of its flow. In other words, if a velocity field $\vec{V}$ has to exist in a fluid continuum, the velocity components must obey the mass conservation principle. Velocity components in accordance with the mass conservation principle are said to constitute a possible fluid flow, whereas these in violation of this principle, are said to constitute an impossible flow. Therefore, the existence of a physically possible flow field is verified from the principle of conservation of mass.

In Section (3.3.5), we have shown that condition for incompressibility of fluid flow leads to the kinematic constraint $\nabla.\vec{V} = 0$. Interestingly, this consideration is intimately related to the consideration of the conservation of mass for incompressible flows. In this section, we will emphasise on the mathematical description of conservation of mass for general fluid flows from different perspectives, and eventually arrive at certain specific conclusions related to the kinematic constraints of incompressible flows.

3.4.1 Conservation of Mass for a Fluid Element (Lagrangian Point of View: A Control Mass System Approach)

Let us consider a fluid element of mass m, density ρ, and volume $\forall$. Mass of a fluid element can be expressed in terms of elemental density and elemental volume as

$$m = \rho \forall$$

or

$$\text{In } m = \text{In } \rho + \text{In } \forall$$

Differentiating (obtaining total derivative) with respect to time t, we obtain

$$\frac{1}{m}\frac{Dm}{Dt} = \frac{1}{\rho}\frac{D\rho}{Dt} + \frac{1}{\forall}\frac{D\forall}{Dt}$$

Since mass of the element is conserved, $\frac{Dm}{Dt} = 0$, which implies that we can write in a Cartesian system (noting that $\frac{1}{\forall}\frac{D\forall}{Dt}$ is the rate of volumetric strain; see Eq. 3.24d)

$$0 = \frac{1}{\rho}\frac{D\rho}{Dt} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

$$\Rightarrow \quad 0 = \frac{1}{\rho}\left[\frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z}\right] + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

$$\Rightarrow \quad \frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z} + \rho\frac{\partial u}{\partial x} + \rho\frac{\partial v}{\partial y} + \rho\frac{\partial w}{\partial z} = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \qquad (3.45)$$

Equation. (3.45) is the well-known equation of continuity in a rectangular Cartesian coordinate system. The equation can be written in a vector form as

$$\frac{\partial \rho}{\partial t} + \nabla.\left(\rho \vec{V}\right) = 0 \tag{3.46}$$

where $\vec{V}$ represents the velocity vector. Equation. (3.46) is the celebrated continuity equation in fluid mechanics, which mathematically represents the conservating mass for fluid flow.

In case of steady flow,

$$\frac{\partial \rho}{\partial t} = 0$$

Hence, Eq. (3.46) becomes

$$\nabla.\left(\rho \vec{V}\right) = 0 \tag{3.47}$$

or, in a rectangular Cartesian coordinate system

$$\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \tag{3.48}$$

In case of incompressible flow, we have earlier seen by definition that

$$\nabla.\vec{V} = 0 \tag{3.49}$$

or, in a Cartesian system $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ (3.50)

Substituting Eq. (3.49) in Eq. (3.46), we thus get, for incompressible flow

$$\frac{\partial \rho}{\partial t} + \vec{V}.\nabla\rho = 0 \tag{3.51}$$

or $$\frac{D\rho}{Dt} = 0 \tag{3.51a}$$

Equation. (3.51a) implies that for an incompressible flow, the total derivative of density is zero (in a Cartesian system: $\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z} = 0$). *However, this does not necessarily imply that the density has to be a constant for incompressible flow.* In fact, the density can be functions of position and time in Eq. (3.50) in a manner such that all terms in the expression for $\frac{D\rho}{Dt}$ cancel each other, leading to $\frac{D\rho}{Dt} = 0$. Such flows are known as variable density, incompressible flows. Nevertheless, for the special case of a constant density flow, $\frac{D\rho}{Dt}$ is trivially zero. Thus, constant density flow is a special outclass of incompressible flow.

3.4.2 Alternate Derivation of Continuity Equation (Eulerian Point of View: A Control Volume Approach)

In order to derive the continuity equation from an Eulerian perspective, one may consider a differentially small rectangular control volume (i.e., an identified volume in space across which fluid flows) with coordinate axes as shown in Fig. 3.15. Importantly, fluid velocity components normal to the faces of the control volume solely contribute towards the volume flow rates across those respective faces. For illustration, we consider first the flow along the x direction, and subsequently extend similar inferences to y and z directions. Let the fluid enter across the face ABCD with a velocity u and density ρ. Therefore, the rate of mass entering the control volume (CV) along the x direction through face ABCD is given by

$$(\dot{m}_{in})_x = \rho u \Delta y \Delta z$$

The rate of mass leaving the control volume along the x direction through face EFGH is

$$(\dot{m}_{out})_{x+\Delta x} = (\dot{m}_{in})_x + \frac{\partial}{\partial x}(\dot{m}_{in})_x \Delta x + \text{higher order terms (h.o.t.)}$$

$$= \rho u \Delta y \Delta z + \frac{\partial}{\partial x}(\rho u)\Delta x \Delta y \Delta z + \text{h.o.t.}$$

Hence, the net rate of mass entering the control volume along the x direction is

$$(\dot{m}_{in})_x - (\dot{m}_{out})_{x+\Delta x} = -\frac{\partial}{\partial x}(\rho u)\Delta x \Delta y \Delta z + \text{h.o.t.}$$

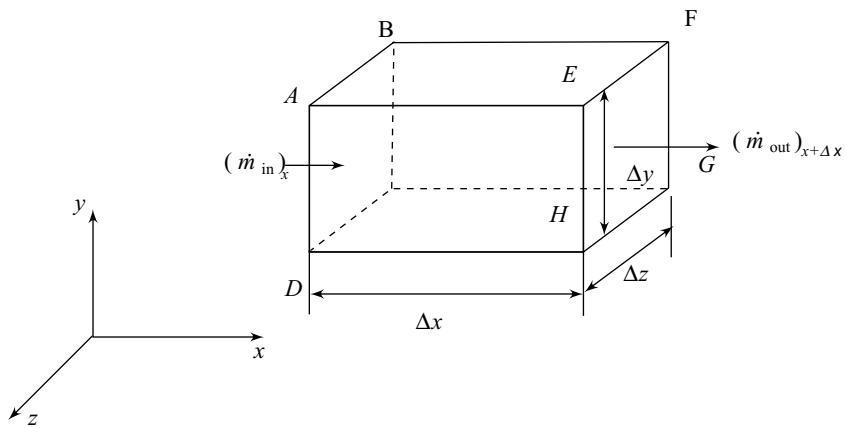


Fig. 3.15

In a similar fashion, the net rate of mass entering the control volume along the y direction is

$$(\dot{m}_{in})_y - (\dot{m}_{out})_{y+\Delta y} = -\frac{\partial}{\partial y}(\rho v)\Delta x\Delta y\Delta z + \text{h.o.t.}$$

and the net rate of mass entering the control volume along the z direction is

$$(\dot{m}_{in})_z - (\dot{m}_{out})_{z+\Delta z} = -\frac{\partial}{\partial z}(\rho w)\Delta x\Delta y\Delta z + \text{h.o.t.}$$

The conservation of mass for the control volume gives

$$\dot{m}_{in} - \dot{m}_{out} = \frac{\partial}{\partial t}(m_{CV}) = \frac{\partial}{\partial t}(\rho\Delta x\Delta y\Delta z) = \frac{\partial \rho}{\partial t}\Delta x\Delta y\Delta z \tag{3.52}$$

(Since $\Delta x\Delta y\Delta z$ is invariant with time).
Taking $\Delta x, \Delta y, \Delta z \to 0$ so that higher order term $\to 0$, and substituting expressions for $\dot{m}_{in} - \dot{m}_{out}$ different terms in Eq. (3.52), we obtain

$$-\frac{\partial}{\partial x}(\rho u)\Delta x\Delta y\Delta z - \frac{\partial}{\partial y}(\rho v)\Delta x\Delta y\Delta z - \frac{\partial}{\partial z}(\rho w)\Delta x\Delta y\Delta z = \frac{\partial \rho}{\partial t}\Delta x\Delta y\Delta z$$

$$\left[\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w)\right]\Delta x\Delta y\Delta z = 0$$

Since the equation is valid irrespective of the size $\Delta x\ \Delta y\ \Delta z$ of the control volume, we can write

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

or,
$$\frac{\partial \rho}{\partial t} + \nabla.\left(\rho \vec{V}\right) = 0$$

3.4.3 One-dimensional Cross-sectionally Averaged Form of the Continuity Equation

Consider steady flow through a variable area conduit, as shown in Fig. 3.16(a).

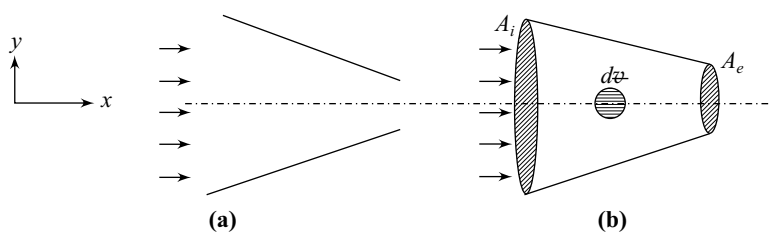


Fig. 3.16

The corresponding continuity equation reads

$$\nabla.\left(\rho\vec{V}\right)=0$$

Integrating the same over the domain (an elemental volume $d\forall$ is depicted in Fig. 3.16b), we get

$$\int_{\forall}\nabla.\left(\rho\vec{V}\right)d\forall=0$$

Using the divergence theorem, the above integral may be converted into an integral over an area that bounds the volume $\forall$, so that one may write

$$\int_{A}\left(\rho\vec{V}\right).\hat{n}dA=0$$

where $\hat{n}$ is a unit vector normal to dA. Since the velocity vector is zero except for the inflow (A_i) and outflow (A_e) boundaries, the above integral may be written as

$$\int_{A_i}\left(\rho\vec{V}\right).\hat{n}dA+\int_{A_e}\left(\rho\vec{V}\right).\hat{n}dA=0$$

Since $\hat{n}=-\hat{i}$ over A_i and $\hat{n}=\hat{i}$ over A_e, one may write the above as

$$-\int_{A_i}\rho udA+\int_{A_e}\rho udA=0 \tag{3.53}$$

where u is the x component of flow velocity.

For steady flow, thus the mass flow rate at section i is equal to the mass flow rate at section e. Many times it is convenient to express the above in terms of average velocity. For example, consider a situation in which density does not vary over a given section. Further, we note that the cross-sectionally average velocity is defined as

$$\bar{u}=\frac{\int_{A}udA}{A} \tag{3.54}$$

Physically, the average velocity is an equivalent uniform velocity that could have given rise to the same volumetric flow rate as that induced by the variable velocity under consideration. Combining Eqs (3.53) and (3.54), one can write

$$\rho_i\bar{u}_iA_i=\rho_e\bar{u}_eA_e \tag{3.55}$$

where the subscript i and e indicate the inlet and exit conditions respectively.
If density is not varying spatially, i.e., $\rho_i=\rho_e$, Eq. (3.55) becomes

$$\bar{u}_iA_i=\bar{u}_eA_e \tag{3.56}$$

3.4.4 Continuity Equation in a Cylindrical Polar Coordinate System

The continuity equation in any coordinate system can be derived in two ways, viz., (i) either by expanding the vectorial form of general Eq. (3.46) with respect to the particular coordinate system, or (ii) by considering an elemental control volume appropriate to the reference frame of coordinates and then by applying the

fundamental principle of conservation of mass. The term $\nabla\cdot(\rho\vec{V})$ in a cylindrical polar coordinate system (Fig. 3.17) can be written as

$$\nabla\cdot(\rho\vec{V}) = \frac{\partial}{\partial r}(\rho v_r) + \frac{\rho v_r}{r} + \frac{1}{r}\frac{\partial(\rho v_\theta)}{\partial\theta} + \frac{\partial}{\partial z}(\rho v_z) \tag{3.57}$$

Therefore, the equation of continuity in a cylindrical polar coordinate system can be written as

$$\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial r}(\rho v_r) + \frac{\rho v_r}{r} + \frac{1}{r}\frac{\partial(\rho v_\theta)}{\partial\theta} + \frac{\partial}{\partial z}(\rho v_z) = 0 \tag{3.58}$$

The above equation can also be derived by considering the mass fluxes in the control volume shown in Fig. 3.18.

Rate of mass entering the control volume through face $ABCD$

$$= \rho\, v_r\, r\, \mathrm{d}\theta\, \mathrm{d}z$$

Rate of mass leaving the control volume through the face $EFGH$

$$= \rho\, v_r\, r\, \mathrm{d}\theta\, \mathrm{d}z + \frac{\partial}{\partial r}(\rho v_r r\, \mathrm{d}\theta\, \mathrm{d}z)\mathrm{d}r$$

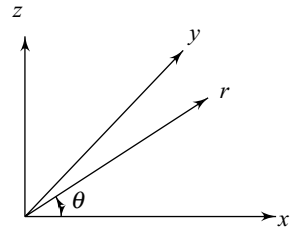


Fig. 3.17 A cylindeical polar coordinate system

Hence, the net rate of mass efflux in the r direction = $\frac{1}{r}\frac{\partial}{\partial r}(\rho v_r\, r)\,\mathrm{d}\forall$

where, $\mathrm{d}\forall = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$ (the elemental volume)

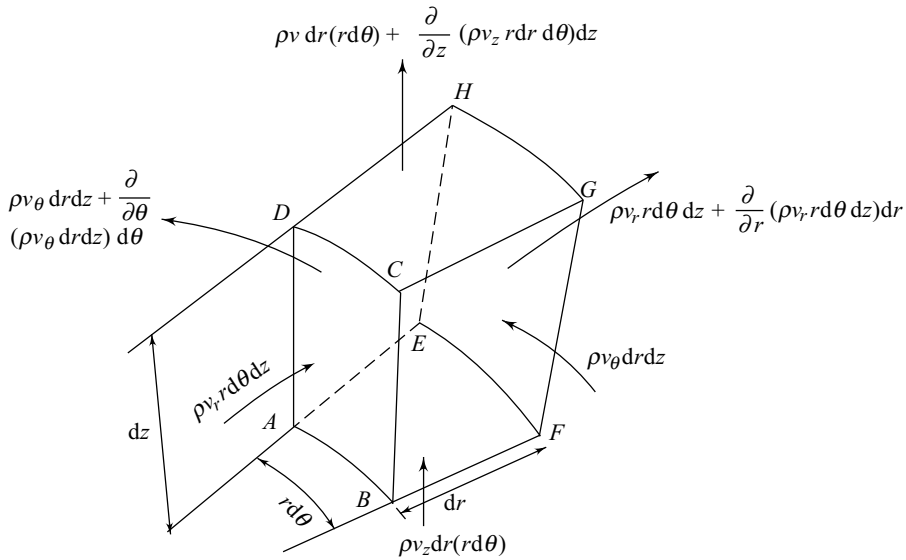


Fig. 3.18 A control volume appropriate to a cylindrical polar coordinate system

The net rate of mass efflux from the control volume, in θ direction, is the difference of mass leaving through face *ADHE* and the mass entering through face *BCGF* and can be written as

$$\frac{1}{r}\frac{\partial}{\partial\theta}(\rho v_\theta)\,\mathrm{d}\forall\,.$$

The net rate of mass efflux in the z direction can be written in a similar fashion as

$$\frac{\partial}{\partial z}(\rho v_z)\,\mathrm{d}\forall$$

The rate of increase of mass within the control volume becomes

$$\frac{\partial}{\partial t}(\rho\,\mathrm{d}\forall) = \frac{\partial\rho}{\partial t}(\mathrm{d}\forall)$$

Hence, following the fundamental principle of conservation of mass (rate of accumulation of mass in the control volume + net rate of mass efflux from the control volume = 0), the final form of continuity equation in a cylindrical polar coordinate system becomes

$$\frac{\partial\rho}{\partial t}+\frac{1}{r}\frac{\partial}{\partial r}(\rho v_r\, r)+\frac{1}{r}\frac{\partial}{\partial\theta}(\rho v_\theta)+\frac{\partial}{\partial z}(\rho v_z)=0$$

or,
$$\frac{\partial\rho}{\partial t}+\frac{\partial}{\partial r}(\rho v_r)+\frac{\rho v_r}{r}+\frac{1}{r}\frac{\partial}{\partial\theta}(\rho v_\theta)+\frac{\partial}{\partial z}(\rho v_z)=0$$

In case of an incompressible flow,

$$\frac{\partial v_r}{\partial r}+\frac{v_r}{r}+\frac{1}{r}\frac{\partial v_\theta}{\partial\theta}+\frac{\partial v_z}{\partial z}=0 \tag{3.59}$$

The equation of continuity in a spherical polar coordinate system (Fig. 3.1) can be written by expanding the term $\nabla\cdot(\rho\vec{V})$ of Eq. (3.46) as

$$\frac{\partial\rho}{\partial t}+\frac{1}{R^2}\frac{\partial}{\partial R}(R^2\,\rho v_R)+\frac{1}{R\sin\phi}\frac{\partial(\rho v_\theta)}{\partial\theta}+\frac{1}{R\sin\phi}\frac{\partial(\rho v_\phi\sin\phi)}{\partial\phi}=0 \tag{3.60}$$

For an incompressible flow, Eq. (3.60) reduces to

$$\frac{1}{R}\frac{\partial}{\partial R}(R^2 v_R)+\frac{1}{\sin\phi}\frac{\partial v_\theta}{\partial\theta}+\frac{1}{\sin\phi}\frac{\partial(v\phi\sin\phi)}{\partial\phi}=0 \tag{3.61}$$

The derivation of Eq. (3.60) by considering an elemental control volume appropriate to a spherical polar coordinate system is left as an exercise for the readers.

Example 3.13

Does a velocity field given by

$$\vec{V} = 5x^3\,\hat{i}-15x^2\,y\,\hat{j}+t\,\hat{k}$$

represent a possible incompressible flow of fluid?

Solution

In order to check for a physically possible incompressible fluid flow, one has to look for its compliance with the equation of continuity.

The continuity equation (in differential form) for a three-dimensional incompressible flow can be written as

$$\nabla \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Here, $u = 5x^3, v = -15x^2y$ and $w = t,$

Hence, $\frac{\partial u}{\partial x} = 15x^2, \frac{\partial \mathrm{v}}{\partial y} = -15x^2$ and $\frac{\partial w}{\partial z} = 0$

which, on substitution in the continuity equation satisfies it for all x, y, z and t values. This shows that the above velocity field represents a physically possible incompressible flow.

Example 3.14

Which of the following sets of equations represent possible two-dimensional incompressible flows?

(i) $u = x + y; \quad v = x - y$
(ii) $u = x + 2y; \quad v = x^2 - y^2$
(iii) $u = 4x + y; \quad v = x - y^2$
(iv) $u = xt + 2y; \quad v = x^2 - yt^2$
(v) $u = xt^2; \quad v = xyt + y^2$

Solution

The continuity equation (in differential form) for a two-dimensional incompressible flow is $\frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 0$

(i) $u = x + y \quad v = x - y; \quad \frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 1 + (-1) = 0$

$\therefore$ two-dimensional incompressible flow is possible.

(ii) $u = x + 2y, \quad v = x^2 - y^2; \quad \frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 1 - 2y$

$\therefore$ two-dimensional incompressible flow is not possible.

(iii) $u = 4x + y, \quad v = x - y^2; \quad \frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 4 - 2y$

$\therefore$ two-dimensional incompressible flow is not possible.

(iv) $u = xt + 2y,\ v = x^2 - yt^2;\quad \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = t - t^2$

$\therefore$ two-dimensional incompressible flow is not possible.

(v) $u = xt^2, v = xyt + y^2;\ \dfrac{\partial u}{\partial x} + \dfrac{\partial \mathrm{v}}{\partial y} = t^2 + xt + 2y$

two-dimensional incompressible flow is not possible.

Example 3.15

For a flow in the xy plane, the y component of velocity is given by

$$v = y^2 - 2x + 2y$$

Determine a possible x component for a steady incompressible flow. How many possible x components are there?

Solution

The flow field is steady and incompressible. Therefore, from continuity

$$\frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 0$$

or

$$\frac{\partial u}{\partial x} = -\frac{\partial \mathrm{v}}{\partial y}$$

Now,

$$-\frac{\partial \mathrm{v}}{\partial y} = -\frac{\partial}{\partial y}(y^2 - 2x + 2y) = -(2y + 2) = -2y - 2$$

Hence,

$$u = \int \frac{\partial u}{\partial x}\,\mathrm{d}x = \int -\frac{\partial \mathrm{v}}{\partial y}\,\mathrm{d}x = \int -(2y + 2)\,\mathrm{d}x$$

$$= -2yx - 2x + f(y)$$

There are infinite number of possible x components, since $f(y)$ is arbitrary. The simplest one would be found by setting $f(y) = 0$.

Example 3.16

A heated rectangular electronic chip floats on the top of a thin layer of air, above a bottom plate. Air is blown at a uniform velocity v_0 through holes in the bottom plate. For steady, inviscid, and constant density flow, find out the components of velocity and acceleration. Width of the chip perpendicular to the plane of the figure is b.

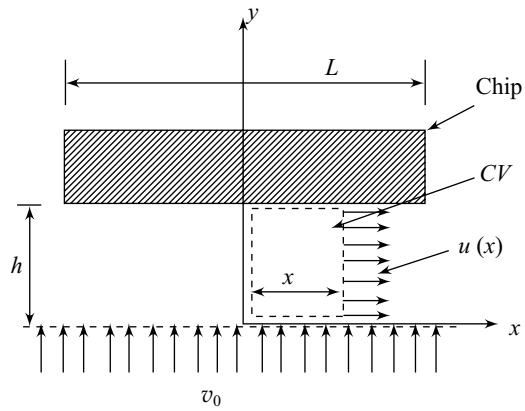


Fig. 3.19

Solution

Choose a fixed control volume as shown by the dashed line in Fig. 3.19.
The rate at which air enters this control volume through the bottom is the same rate at which it leaves the control volume across the gap, to maintain continuity. Thus, one may write

$$v_0\, xb = u(x)\, hb$$

or $$u(x) = \frac{v_0 x}{h}$$

Continuity equation becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

or $$\frac{v_0}{h} + \frac{\partial v}{\partial y} = 0$$

or $$\partial v = -\frac{v_0}{h}\partial y$$

$$\therefore v = -\frac{v_0}{h} y + f(x) \tag{3.62}$$

The integration constant $f(x)$ can be found by applying an appropriate boundary condition as follows:

$$\text{At } y = h, \quad v = 0$$

$$\therefore f(x) = \frac{v_0}{h} h = v_0$$

Thus, $$v = v_0\left(1 - \frac{y}{h}\right) \tag{3.62a}$$

$$a_x = \cancel{\frac{\partial u}{\partial t}}^{0} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}$$

$$= \frac{v_0 x}{h}\frac{v_0}{h} = \frac{{v_0}^2 x}{h^2}$$

$$a_y = \cancel{\frac{\partial v}{\partial t}}^{0} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}$$

$$= v_0\left(1-\frac{y}{h}\right)\left(-\frac{v_0}{h}\right) = -\frac{{v_0}^2}{h}\left(1-\frac{y}{h}\right)$$

The acceleration field can be expressed as

$$\vec{a} = \frac{{v_0}^2 x}{h^2}\hat{i} - \frac{{v_0}^2}{h}\left(1-\frac{y}{h}\right)\hat{j}$$

This example illustrates that although velocity is not a function of time but still there is an acceleration, because of the convective component (originating out of spatial gradients of flow velocities).

Example 3.17

Fluid flows through a convergent nozzle as shown in Fig. 3.20. Cross-sectional area of the nozzle is given as $A = A_0(1 - bx)$, where x is measured from the entrance, A_0 is the cross-sectional area of the nozzle at the entrance, and b is a constant. The free stream velocity is given as $u_\infty = c(1 + at)$, where c and a are dimensional constants. Find out an expression for acceleration assuming inviscid flow.

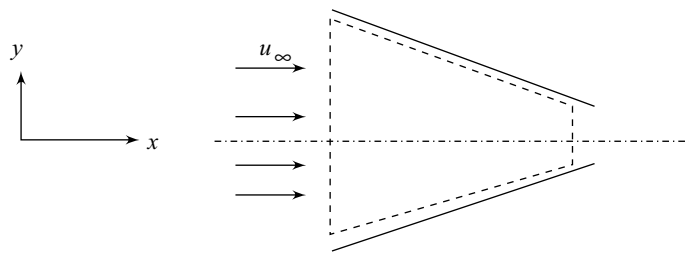


Fig. 3.20

Solution

The cross-sectionally averaged continuity equation provides us with

$$u_\infty A_0 = u(x,t)A(x)$$

or

$$u(x,t) = \frac{c(1+at)A_0}{A_0(1-bx)} = \frac{c(1+at)}{(1-bx)}$$

The acceleration field can be expressed as

$$a_x = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{ca}{(1-bx)} + \frac{c(1+at)}{(1-bx)}\frac{cb(1+at)}{(1-bx)^2} = \frac{c}{(1-bx)}\left[a + \frac{cb(1+at)^2}{(1-bx)^2}\right]$$

Example 3.18

The two plates separated by a distance b form a channel (Fig. 3.21). One of the plates is porus and the other one is impermeable. A flow takes place within the channel so that the x component velocity u is a function of x only and its value at the inlet is u_0. There is a uniform inflow v_0 through the porus wall to the channel so that the velocity component v in the y direction within the channel is a function of y only. Considering the flow to be incompressible, find the expression of u as a function of x, and v as a function of y.

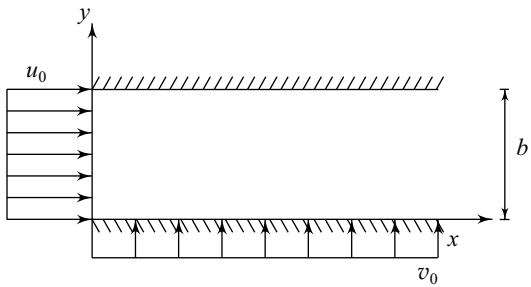


Fig. 3.21 Flow through a channel formed by two plates

Solution

The equation of continuity at any point within the channel can be written as

$$\frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 0$$

or
$$\frac{\partial u}{\partial x} = -\frac{\partial \mathrm{v}}{\partial y} \tag{3.63}$$

Since u is a function of x only and v is a function of y only, the equality of their derivatives, as expressed by the Eq. (3.63), would be valid provided both of them are equal to a constant. Hence, we can write

$$\frac{du}{dx} = -\frac{d\mathrm{v}}{dy} = K \text{ (a constant)}$$

which give
$$\frac{du}{dx} = K \tag{3.64a}$$

$$\frac{d\mathrm{v}}{dy} = -K \tag{3.64b}$$

Integration of Eqs (3.64a) and (3.64b) gives

$$u = Kx + C_1 \tag{3.65a}$$

$$v = -Ky + C_2 \tag{3.65b}$$

Using the boundary conditions

$$\text{at } x = 0 \qquad u = u_0$$
$$\text{at } y = 0 \qquad v = v_0$$

and at $y = b$ $v = 0$

in Eqs (3.65a) and (3.65b), we have

$$C_1 = u_0,\ C_2 = v_0,\ K = v_0/b$$

Substituting the values of K, C_1 and C_2 in Eqs (3.65a) and (3.65b), the final expressions for u and v are obtained as

$$u = u_0 + \frac{v_0}{b}x \qquad \text{and} \qquad v = v_0 - \frac{v_0}{b}y$$

3.5 STREAM FUNCTION

The concept of stream function is a direct consequence of the principle of continuity. Let us consider a two-dimensional incompressible flow parallel to the xy plane in a rectangular Cartesian coordinate system. The flow field in this case is defined by

$$u = u(x, y, t)$$
$$v = v(x, y, t)$$
$$w = 0$$

The equation of continuity is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3.66}$$

If a function $\psi(x, y, t)$ is defined in the manner

$$u = \frac{\partial \psi}{\partial y} \tag{3.67a}$$

and

$$v = -\frac{\partial \psi}{\partial x} \tag{3.67b}$$

so that it automatically satisfies the equation of continuity (Eq. (3.66)), then the *function* ψ is known as *stream function*. For a steady flow, ψ is a function of two variables x and y only. In case of a two-dimensional irrotational flow,

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

so that

$$\frac{\partial}{\partial x}\left(-\frac{\partial \psi}{\partial x}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial y}\right) = 0$$

or

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0 \tag{3.68}$$

Thus, for an irrotational flow, stream function satisfies the *Laplace's equation*.

3.5.1 Constancy of ψ on a Streamline

Since ψ is a point function, it has a value at every point in the flow field. Hence, a change in the stream function ψ can be written as

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v\, dx + u\, dy$$

Further, the equation of a streamline is given by

$$\frac{u}{dx} = \frac{v}{dy} \qquad \text{or} \qquad u\, dy - v\, dx = 0$$

It follows that $d\psi = 0$ on a streamline, i.e., the value of ψ is constant along a streamline. Therefore, the equation of a streamline can be expressed in terms of the stream function as

$$\psi(x, y) = \text{constant} \tag{3.69}$$

Once the function ψ is known, streamline can be drawn by joining the same values of ψ in the flow field.

3.5.2 Physical Significance of Stream Function

Figure 3.22(a) illustrates a two-dimensional flow. Let A be a fixed point, but P be any point in the plane of the flow. The points A and P are joined by the arbitrary lines ABP and ACP. For an incompressible steady flow, the volume flow rate across ABP into the space $ABPCA$ (considering a unit width in a direction perpendicular to the plane of the flow) must be equal to that across ACP. A number of different paths connecting A and P (ADP, AEP, ...) may be imagined but the volume flow rate across all the paths would be the same. This implies that the rate of flow across any curve between A and P depends only on the end points A and P.

Since A is fixed, the rate of flow across ABP, ACP, ADP, AEP (any path connecting A and P) is a function only of the position P. This function is known as the *stream function* ψ. The value of ψ at P therefore represents the volume flow rate across any line joining P to A. The value of ψ at A is made arbitrarily zero. The fixed point A may be the origin of coordinates, but this is not necessary. If a point P' is considered (Fig. 3.22(b)), PP' being along a streamline, then the rate of flow across the curve joining A to P' must be the same as across AP, since, by the definition of a streamline, there is no flow across PP'. The value of ψ thus remains same at P' and P. Since P' was taken as any point on the streamline through P, it follows that ψ is constant along a streamline. Thus the flow may be represented by a series of streamlines at equal increments of ψ. If another point P'' is considered (Fig. 3.22(b)) in the plane, such that PP'' is a small distance δn perpendicular to the streamline through P with $AP'' > AP$, then the volume flow rate across the curve AP'' is greater than that across AP by the increment $\delta\psi$ of the stream function from point P to P''. Let the average velocity perpendicular to PP'' (i.e., in the direction of streamline at P) be V, then

$$\delta\psi = V \cdot \delta n$$

or

$$V = \delta\psi / \delta n$$

Therefore, the velocity at a point can be expressed in terms of the stream function ψ as

$$V = \lim_{\delta n \to 0} \frac{\delta \psi}{\delta n} = \frac{\partial \psi}{\partial n}$$

This gives the mathematical definition of the stream function at the point *P*. The above concept can be visualised more easily by considering the flow between two adjacent streamlines in a rectangular Cartesian coordinate system (Fig. 3.22(c)). Let the values of the stream functions for the two streamlines be denoted by ψ and $\psi + d\psi$. The volume flow rate dQ for an incompressible flow across any line, say AB, of unit width, joining any two points A and B on two streamlines, can be written as

$$dQ = d\psi$$

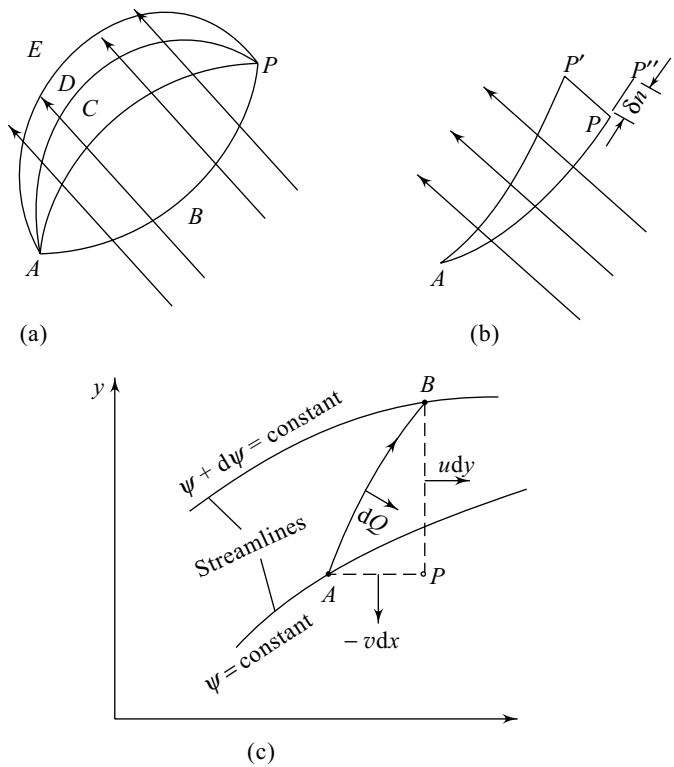


Fig. 3.22 Physical interpretation of stream function

As ψ is a function of space coordinates, x and y,

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

Hence,
$$dQ = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy \tag{3.70}$$

Again, the volume of fluid crossing the surface AB must be flowing out from surfaces AP and BP of unit width. Hence,

$$dQ = u\,dy - v\,dx \tag{3.71}$$

Comparing the Eqs (3.70) and (3.71), we get

$$u = \frac{\partial \psi}{\partial y} \qquad \text{and} \qquad v = -\frac{\partial \psi}{\partial x}$$

The stream function, in a polar coordinate system is defined as

$$v_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} \qquad \text{and} \qquad v_\theta = -\frac{\partial \psi}{\partial r}$$

The expressions for v_r and v_θ in terms of the stream function automatically satisfy the equation of continuity given by

$$\frac{\partial}{\partial r}(v_r\, r) + \frac{\partial}{\partial \theta}(v_\theta) = 0$$

3.5.3 Stream Function in a Three-Dimensional Flow

It is not possible to draw a streamline with a single stream function in case of a three-dimensional flow. An axially symmetric three-dimensional flow is similar to the two-dimensional case in a sense that the flow field is the same in every plane containing the axis of symmetry. The equation of continuity in the cylindrical polar coordinate system for an incompressible flow is given by Eq (3.59). For an axially symmetric flow (the axis $r = 0$ being the axis of symmetry), the simplified form of Eq. (3.59) without the term $\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}$ is satisfied by a function defined as

$$r\, v_r = -\frac{\partial \psi}{\partial z}, \qquad r\, v_z = \frac{\partial \psi}{\partial r} \tag{3.72}$$

The function ψ, defined by the Eq. (3.72) in case of a three-dimensional flow with an axial symmetry, is called the *stokes stream function*.

3.5.4 Stream Function in Compressible Flow

Definition of the stream function ψ for a two-dimensional compressible flow offers no difficulty. Instead of relating it to the volume flow rate, one can relate it to the mass flow rate. The continuity equation for a steady two-dimensional compressible flow is given by

$$\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) = 0$$

Hence a stream function ψ can be defined which will satisfy the above equation of continuity as

$$\rho\, u = \rho_0 \frac{\partial \psi}{\partial y}$$

$$\rho\, v = -\rho_0 \frac{\partial \psi}{\partial x} \tag{3.73}$$

where ρ_0 is a reference density and is used in the manner as shown in Eq. (3.73), to retain the unit of ψ same as that in the case of an incompressible flow. Therefore, from a physical point of view, the difference in stream function between any two streamlines multiplied by the reference density ρ_0 will give the mass flow rate through the passage of unit width formed by the streamlines.

Example 3.19

A stream function is given by

$$\psi = 2x^2y + (3 + t)y^2$$

Find the flow rates across the faces of the triangular prism OAB, shown in Fig. 3.23, having a thickness of 1 unit in the z direction at time $t = 1$.

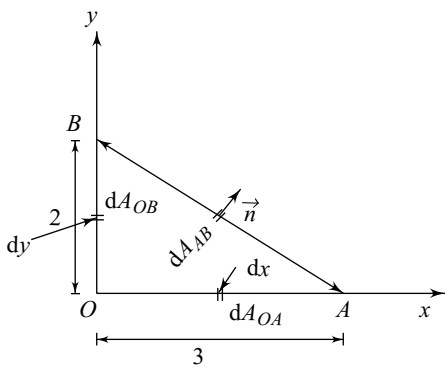


Fig. 3.23 The faces of a triangular prism

Solution

The velocity field corresponding to the given stream function can be written according to Eqs (3.67a) and (3.67b) as

$$u = \frac{\partial \psi}{\partial y} = 2x^2 + 2(3 + t)y$$

$$v = -\frac{\partial \psi}{\partial x} = -4xy$$

where u and v are the velocity components along the x and y directions, respectively.

At $t = 1$

$$(u)_{t=1} = 2x^2 + 8y$$

$$(v)_{t=1} = -4xy$$

The volume flow rate across the face perpendicular to the x direction and with the edge OB as seen in the xy plane is found as

$$Q_{OB} = \int_{A_{OB}} (u)_{\substack{t=1 \\ x=0}} \, dA_{OB} = \int_0^2 8y \cdot dy = 16 \text{ units}$$

Similarly, the flow rate across the face with edge OA (as seen in the xy plane) and perpendicular to the y direction becomes

$$Q_{OA} = \int_{A_{OA}} (v)_{\substack{t=1 \\ y=0}} \, dA_{OA}$$

$$= 0$$

Since the z component of velocity is zero, the volume flow rates across the faces perpendicular to the z direction (i.e., face OAB and the face parallel to it and separated by a unit distance) become zero.

Volume flow rate across the inclined face with AB as the edge seen on the xy plane can be written as

$$Q_{AB} = \int \hat{n} \, dA_{AB} \cdot \vec{V} \tag{3.74}$$

where $\vec{n}$ is the unit vector along the normal to the element of surface dA_{AB}, taken positive when directed outwards as shown in Fig. 3.23. Hence, we can write Eq. (3.74) as

$$Q_{AB} = \int_{A_{AB}} [\hat{i}\, dy + \hat{j}\, dx] \cdot [\hat{i}(2x^2 + 8y) + \hat{j}(-4xy)]$$

(where, $\hat{i}$ and $\hat{j}$ are the unit vectors along the x and y directions, respectively)

$$= \int_{A_{AB}} [2x^2 + 8y] dy + \int_{A_{AB}} (-4xy)\, dx \tag{3.75}$$

Again from the geometry,

$$\frac{y}{3-x} = \frac{2}{3}$$

along the surface AB.

Using the relation in Eq. (3.75), we get

$$Q_{AB} = \int_0^2 \left[2\left(3 - \frac{3}{2}y\right)^2 + 8y \right] dy - \int_0^3 4\left(2 - \frac{2}{3}x\right) x \, dx = 16 \text{ units}$$

Example 3.20

An incompressible flow around a circular cylinder of radius a, is represented by the stream function

$$\psi = -U r \sin\theta + \frac{U a^2 \sin\theta}{r}$$

where U represents the free stream velocity. Show that v_r (the radial component of velocity) = 0 along the circle, $r = a$. Find the values of θ at $r = a$, where $|\vec{V}| = U$.

Solution

In a polar coordinate system,

$$v_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta}, \qquad v_\theta = -\frac{\partial\psi}{\partial r}$$

So $$v_r = \frac{1}{r}\left[-U r\cos\theta + \frac{Ua^2}{r}\cos\theta\right] = -U\cos\theta\left(1 - \frac{a^2}{r^2}\right)$$

and $$v_\theta = \left[U\sin\theta + \frac{Ua^2}{r^2}\sin\theta\right] = U\sin\theta\left(1 + \frac{a^2}{r^2}\right)$$

at $r = a$, $v_r = 0$ for all values of θ and $v_\theta = 2U\sin\theta$

Therefore, along the circle $r = a$, $|\vec{V}| = |v_\theta| = |2U\sin\theta|$

Putting $|\vec{V}| = U$, we get $\sin\theta = \pm\frac{1}{2}$, i.e., when $\theta = +30°, 150°, 210°$ and $330°$.

Example 3.21

The velocity components for a steady flow are given as

$u = 0$, $v = -y^3 - 4z$, $w = 3y^2 z$. Determine: (i) whether the flow field is one-,two-, or three-dimensional, (ii) whether the flow is incompressible or compressible, and (iii) the stream function for the flow.

Solution

(i) Since the velocity field is a function of y and z only, the flow field is two-dimensional.

(ii) For an incompressible flow the continuity equation can be written in differential form as

$$\nabla.\vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Here, $$u = 0, v = -y^3 - 4z \text{ and } w = 3y^2 z$$

Hence, $$\frac{\partial u}{\partial x} = 0 \quad \frac{\partial v}{\partial y} = -3y^2 \quad \text{and} \quad \frac{\partial w}{\partial z} = 3y^2$$

which, on substitution in the continuity equation satisfies it for all x, y, z and t values. This shows that the above velocity field represents a physically possible incompressible flow.

(iii) From the definition of stream function ψ,

$$v = \frac{\partial \psi}{\partial z}$$

$$\psi = \int v dz = -y^3 z - 2z^2 + f(y) + C_1$$

$$w = -\frac{\partial \psi}{\partial y}$$

$$\psi = -\int w dy = -y^3 z + g(z) + C_2$$

Comparing the two expressions for ψ, we find

$$f(y) = 0, g(z) = -2z^2$$

Hence, $$\psi = -y^3 z - 2z^2 + C$$

where, C is a constant.

It is important to note that solid wall itself is a stream function, because there is no flow across it. By physical sense, solid boundary of any shape is a streamline. We can give a particular value of stream function if the flow is two-dimensional and incompressible. Although classically we give $\psi = 0$ at the wall but we can give any value. It is just some sort of reference with respect to which one can find the other stream functions.

3.6 VELOCITY POTENTIAL

If $\nabla \times \vec{V} = 0$, then the velocity field is called the *conservative field* and $\vec{V}$ may be expressed in the form of gradient of a scalar function known as *velocity potential.* Thus, in that case,

$$\vec{V} = \nabla \phi \tag{3.76}$$

which implies

$$u = \frac{\partial \phi}{\partial x} \tag{3.76a}$$

$$v = \frac{\partial \phi}{\partial y} \tag{3.76b}$$

$$w = \frac{\partial \phi}{\partial z} \tag{3.76c}$$

In case of a two-dimensional incompressible flow, the continuity equation becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Substituting u, v in terms of ϕ, we obtain

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}=0 \tag{3.77}$$

Thus, velocity potential automatically satisfies the Laplace's equation.

3.6.1 Relationship between Velocity Potential and Stream Function

Consider a two-dimensional, incompressible and irrotational flow so that both the stream function and velocity potential exist.

Since the flow is two-dimensional, $\phi = \phi(x, y)$

Thus,
$$d\phi = \frac{\partial\phi}{\partial x}dx+\frac{\partial\phi}{\partial y}dy$$
$$= udx + vdy$$

For constant ϕ, $d\phi$, i.e.,

$$\left.\frac{dy}{dx}\right|_{\phi=\text{cont}} = -\frac{u}{v} \tag{3.78}$$

Again for two-dimensional, incompressible flow, the stream function is expressed as

$$\psi = \psi(x, y)$$
$$d\psi = \frac{\partial\psi}{\partial x}dx+\frac{\partial\psi}{\partial y}dy$$
$$= -vdx + udy$$

For constant ψ, $d\psi$, = 0 i.e.,

$$\left.\frac{dy}{dx}\right|_{\psi=\text{cont}} = \frac{v}{u} \tag{3.79}$$

$$\left(\left.\frac{dy}{dx}\right|_{\phi=\text{const}}\right)\times\left(\left.\frac{dy}{dx}\right|_{\psi=\text{const}}\right) = -1\,(u, v \neq 0) \tag{3.80}$$

Equation. (3.80) implies that the lines of constant ϕ (equipotential line) and lines of constant ψ (streamline) are orthogonal to each other everywhere in the flow field except at certain points where the velocities are zero (stagnation points).

Example 3.22

The velocity potential function for a flow is given by $\phi = x^2 - y^2$. Verify that the flow is incompressible and then determine the stream function for the flow.

Solution

$$\phi = x^2 - y^2$$

From the definition of velocity potential

$$\vec{V} = \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j}$$

Therefore, $\vec{V} = 2x\,\hat{i} - 2y\,\hat{j}, \qquad u = 2x, v = -2y$

Check for incompressible flow, $\frac{\partial u}{\partial x} + \frac{\partial \mathrm{v}}{\partial y} = 2 - 2 = 0$

From the definition of stream function ψ,

$$u = \frac{\partial\psi}{\partial y}, \quad \psi = \int u\mathrm{d}y = 2xy + f(x) + C_1$$

$$v = -\frac{\partial\psi}{\partial x}, \quad \psi = -\int v\,\mathrm{d}x = 2xy + g(y) + C_2$$

comparing the two expressions for ψ, we find

$$f(x) = g(y) = 0$$

Hence, $\psi = 2xy + C$

where, C is a constant.

SUMMARY

- Kinematics of fluid characterises the different types of motion and associated deformation rates of fluid element, without any reference to the forcing parameters.
- Fluid motion is described by two methods, namely, the Lagrangian method and the Eulerian method. In the Lagrangian view, the velocity and other hydrodynamic parameters are specified for particles or elements of given identities, while, in the Eulerian view, these parameters are expressed as functions of location and time. The Lagrangian version of a flow field can be obtained from the integration of the set of equations describing the flow in the Eulerian version.
- A flow is defined to be steady when the flow velocities and fluid properties at any point do not change with time. Flow in which any of these parameters changes with time is termed as unsteady. A flow may appear steady or unsteady depending upon the choice of coordinate axes. A flow is said to be uniform when flow velocities and fluid properties do not change from point to point at any instant of time, or else the flow is non-uniform.
- The total derivative of velocity with respect to time is known as material or substantial acceleration, while the partial derivative of velocity with respect to time for a fixed location is known as temporal acceleration. Material acceleration = Temporal acceleration + Convective acceleration.

- A streamline at any instant of time is an imaginary curve or line in the flow field so that the tangent to the curve at any point represents the direction of the instantaneous velocity at that point. A path line is the trajectory of a fluid particle of a given identity. A streak line at any instant of time is the locus of temporary locations of all particles that have passed through a fixed point in the flow. In a steady flow, the streamlines, path lines, and streak lines are identical.
- Flow parameters, in general, become functions of time and space coordinates. A one-dimensional flow is that in which the flow parameters are functions of time and one space coordinate only.
- A fluid motion consists of translation, rotation, and continuous deformation. In a uniform flow, the fluid elements are simply translated without any deformation or rotation. The deformation and rotation of fluid elements are caused by the variations in velocity components with the space coordinates. The linear deformation or strain rate is defined as the rate of change of length of a linear fluid element per unit original length. The rate of angular deformation at a point is defined as the rate of change of angle between two linear elements at that point which were initially perpendicular to each other. The rotation at a point is defined as the arithmetic mean of the angular velocities of two perpendicular linear meeting at that point. The rotation of a fluid element in the absence of any deformation is known as pure or rigid body rotation.
- Continuity equation mathematically represents conservation of mass for fluid flow. In an incompressible flow field, the velocity vector is divergence-free. Further, for some cases, total derivative of density becomes zero, though that does not necessarily imply that density is a constant. However, a constant density flow is a special case of incompressible flow. The existence of a physically possible flow field is verified from the principle of conservation of mass.
- Vorticity of flow is defined as curl of the velocity vector, and is twice the angular velocity of flow. Circulation of flow is defined as the contour integral of the dot product of velocity with a oriented line element taken along the contour. When the vorticity (or circulation) at all points in the flow field is zero, the flow is said to be irrotational.
- The concept of stream function is a consequence of continuity. In a two-dimensional incompressible flow, the difference in stream functions between two points gives the volume flow rate (per unit width in a direction perpendicular to the plane of flow) across any line joining the points. The value of stream function is constant along a streamline.
- Irrotationality leads to the condition $\nabla \times \vec{V} = 0$ which demands $\vec{V} = \nabla \phi$, where ϕ is known as a potential function. Equipotential lines and streamlines are orthogonal to each other everywhere in the flow field except at the stagnation points.